

Chapter 1 Sheaves

1.1 Presheaves and Sheaves

1.1.1 Presheaves

Definition 1.1.1.1 (Axiom of Presheaves) Suppose $\mathcal{C}$ be a category. Then, a presheaf F of $\mathcal{C}$ on X , where X is a topological space, is defined as following:

- For all open sets $U \subseteq X$, $F(U) \in \mathcal{C}$;
- For all $V \subset U \subseteq X$, a morphism

$$\rho_{UV} : F(U) \rightarrow F(V)$$

such that

- For open $U \subseteq X$, $F(U) \in \mathcal{C}$;
- For open $W \subseteq V \subseteq U$,

$$\begin{array}{ccc} & F(V) & \\ \rho_{UV} \nearrow & & \searrow \rho_{VW} \\ F(U) & \xrightarrow{\rho_{UW}} & F(W) \end{array}$$

Especially, $\rho_{UU} = id_U$.

And for $s \in F(U)$:

Definition 1.1.1.2

- – Elements $s \in F(U)$ is called the **section** of F over U ;
- Collection $F(U)$ is called the **space of sections** of F over U ;

Especially, for $U = X$:

- $s \in F(U)$ is the **global section**;
- $F(U)$ is **the space of global sections**;
- Alternative Notation:

$$\Gamma(U, F) := F(U) = H^0(U, F)$$

- The morphisms ρ_{UV} is called **restriction map**, and for $V \subseteq U$,

$$s|_V := \rho_{UV}(s), s \in F(U)$$

Morphisms

Definition 1.1.1.3 Consider two presheaves $\mathcal{F}, \mathcal{G} \in Psh(\mathcal{C})$. Then, the morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a family of morphisms in $\mathcal{C}$

$$\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

such that the following square commutes: For $U, V \subseteq X$ opens, $V \subseteq U$:

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{G}(V) \end{array}$$

Constant Prehseaf

Definition 1.1.1.4 Define: Suppose X be a topological space and let A be a fixed set. Define

- $\mathcal{F}(U) := A, U \in \text{Open}(X)$;
- $\rho_U^V := id_A$.

In general:

Definition 1.1.1.5 Suppose $\mathcal{C}$ be a category which contains the final object $\bullet$. Then define the presheaf as following

$$\mathcal{F}(U) := \bullet \quad \rho_{UU} := id_{\bullet}.$$

It is the final object in $\text{Psh}(\mathcal{C})$.

- **Proof** Consider the following morphism of presheaves $\varphi : \mathcal{F} \rightarrow \bullet$:

$$\varphi(U) : \mathcal{F}(U) \rightarrow \bullet$$

□

Presheaf of Abelian Groups

♠ **Proposition 1.1.1.6** Let X be a topological space. The category of presheaves of sets on X has product

$$F \in \text{PSets}, G \in \text{PSets} \Rightarrow F \times G \in \text{PSets}$$

And $(F \times G)(U) = F(U) \times G(U), U \in \text{Open}(X)$.

- **Proof** Given by:

- Elements $U \mapsto F(U) \times G(U)$;
- Restriction Maps for $V \subseteq U \subseteq X$,

$$\rho_U^V : p_U^V \times q_U^V; p, q \text{ are restriction maps of } F, G$$

Furthermore, we have

$$\text{Mor}(\mathcal{H}, F \times \mathcal{G}) = \text{Mor}(\mathcal{H}, F) \times \text{Mor}(\mathcal{H}, \mathcal{G}) \quad (1.1)$$

□

Definition 1.1.1.7 Define the category, $\text{PAb}(X)$

- Objects: $U \mapsto \mathcal{F}(U) \in \text{Ab}$, restriction maps $\rho_U^V : \mathcal{F}(U) \rightarrow \mathcal{F}(V) \in \text{Mor}(\text{Ab})$

- Families $\mathcal{F}(U) \rightarrow \mathcal{G}(U) \in \text{Mor}(\text{Ab})$;

♠ **Proposition 1.1.1.8** The following types of data of structures are equivalent for $\mathcal{F}$ presheaf of sets:

- Definition 1.1.1.7
- $(\mathcal{F}, +, -, 0)$: Morphisms of Presheaves
 - $+$: $\mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$;
 - $-$: $\mathcal{F} \rightarrow \mathcal{F}$;
 - $0 : * \rightarrow \mathcal{F}$

satisfies the axiom of abelian group.

- Map of presheaves $+$: $\mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$; Maps of presheaves $-$: $\mathcal{F} \rightarrow \mathcal{F}$ and a map $0 : * \rightarrow \mathcal{F}$ such that for each open $U \subseteq X$, the quadruple

$$(\mathcal{F}(U), +, -, 0)$$

is an abelian group.

- A map of presheaves $+$: $\mathcal{F} \times \mathcal{F} \rightarrow \mathcal{F}$ such that for every open $U \subseteq X$, the map $+$: $\mathcal{F}(U) \times \mathcal{F}(U) \rightarrow \mathcal{F}(U)$ defines the structure of an abelian group.

(3) $\Leftrightarrow$ (4) Easy to check.

◦ **Proof**

- (1 $\Rightarrow$ 2) Omitted.
- (2 $\Rightarrow$ 3) Consider $(\mathcal{F}(U), +_U, -_U, 0_U)$ for $U \in \text{Open}(X)$;
- (3 $\Rightarrow$ 1) The restriction maps are group homomorphisms: The following diagram commutes

$$\begin{array}{ccc} \mathcal{F}(U) \times \mathcal{F}(U) & \xrightarrow{+|U} & \mathcal{F}(U) \\ \downarrow \rho_U^V \times \rho_U^V & & \downarrow \rho_U^V \\ \mathcal{F}(V) \times \mathcal{F}(V) & \xrightarrow{+|V} & \mathcal{F}(V) \end{array}$$

□

♣ **Example 1.1.1.9** A construction: Given a family $\{M_x\}_{x \in X}$ of abelian groups, and

$$\mathcal{F}(U) := \bigoplus_{x \in U} M_x$$

And restriction map

$$\rho_U^V : \sum_{x \in U} m_x \mapsto \sum_{x \in V} m_x; V \subseteq U$$

Algebraic Structures

Definition 1.1.1.10 (Algebraic Structure) Suppose:

- $\mathcal{C}$ be a category;
- The functor $F \rightarrow \text{Sets}$ is faithful; (Example: Forgetful Functor)
- For $M \rightarrow M'$, a morphism $F(M) \rightarrow F(M')$ the **underlying map** of sets.
- Map of sets $F(M) \rightarrow F(M')$ is equal to some $F(f)$ for $f \in \mathcal{C}$.

Definition 1.1.1.11 A presheaf objects of $\mathcal{C}$, consists of following datas

- A presheaf of sets $\mathcal{F}$ and
- For every open $U \subseteq X$ a choice of an object $A(U) \in \text{Ob}(\mathcal{C})$;

Subject to the following conditions:

- For every open $U \subseteq X$, the set $\mathcal{F}(U)$ is the underlyng set of $A(U)$;
- For $V \subseteq U \subseteq X$, $\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is a morphism of algebraic structures. In other word: For every $V \subseteq U$ open in X the restriction mapping ρ_V^U is the iamge $F(\alpha_V^U)V$ for some unique morphism

$$\alpha_V^U : A(U) \rightarrow A(V)$$

in the category $\mathcal{C}$. Here $F : \mathcal{C} \rightarrow \text{Sets}$ is a faithful functor, usually the forgetful functor.

Then, we recover the presheaf of setst $(\mathcal{F}, \rho_V^U)$ via the rules

$$\mathcal{F}(A(U)) \quad \rho_V^U = F(\alpha_V^U)$$

Where $(A(-), \mathfrak{K}_V^U)$ is what we define as a presheaf with values in $\mathcal{C}$ on X .

Definition 1.1.1.12 Let X be a topological space, $\mathcal{C}$ be a category

- A **presheaf** $\mathcal{F}$ on X with values in $\mathcal{C}$ is given by rule which assigns to every open $U \subseteq X$ with
 - Objects $\mathcal{F}(U) \in \mathcal{C}$;
 - Morphisms $\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ in $\mathcal{C}$ such that whenever $W \subseteq V \subseteq U$, we have

$$\rho_W^U = \rho_V^U \circ \rho_W^V$$

- A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves with value in $\mathcal{C}$ given by a morphism

$$\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

compatible with restriction maps;

Then:

Definition 1.1.1.13 Let $\mathcal{F}$ be a sheaf on X with values in $\mathcal{C}$. The sheaf of sets

$$U \mapsto F(\mathcal{F}(U))$$

is the underlying presheaf of sets of $\mathcal{F}$. $F : \mathcal{C} \rightarrow \text{Sets}$ is a faithful functor.

Presheaf of Modules

Definition 1.1.1.14 Suppose $\mathcal{O}$ be a presheaf of rings on topological space X :

- A **presheaf** of $\mathcal{O}$ -modules is given by an abelian presheaf $\mathcal{F}$ together with a map of presheaves of sets

$$\mathcal{O} \times \mathcal{F} \rightarrow \mathcal{F}$$

such that for every open $U \subseteq X$, the map

$$\mathcal{O}(U) \times \mathcal{F}(U) \rightarrow \mathcal{F}(U)$$

Gives a structure of $\mathcal{O}(U)$ -module structure on $\mathcal{F}(U)$;

- The **morphism** $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ presheaves of $\mathcal{O}$ -modules is a morphism of abelian presheaves φ , such that the following diagram(module)

$$\begin{array}{ccc} \mathcal{O} \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow \\ \mathcal{O} \times \mathcal{G} & \longrightarrow & \mathcal{G} \end{array}$$

commutes.

- The set of $\mathcal{O}$ -module morphisms as above denoted $Hom_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$;
- The category of presheaves of $\mathcal{O}$ -modules is denoted $PMod(\mathcal{O})$.

Especially, for the morphism of presheaves of rings

$$\varphi : \mathcal{O}_1 \rightarrow \mathcal{O}_2$$

Then, $\mathcal{F} \in PMod(\mathcal{O}_2)$ has a $\mathcal{O}_2$ -module structure

$$\mathcal{O}_1 \times \mathcal{F} \rightarrow \mathcal{O}_2 \times \mathcal{F} \rightarrow \mathcal{F}$$

induces the functor $PMod(\mathcal{O}_2) \rightarrow PMod(\mathcal{O}_1), \mathcal{F} \mapsto \mathcal{F}_{\mathcal{O}_1}$.

$$\begin{array}{ccccc} \mathcal{O}_1 \times \mathcal{F} & \longrightarrow & \mathcal{O}_2 \times \mathcal{F} & \longrightarrow & \mathcal{F} \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{O}_1 \times \mathcal{G} & \longrightarrow & \mathcal{O}_2 \times \mathcal{G} & \longrightarrow & \mathcal{G} \end{array}$$

Definition 1.1.1.15 Define the **tensor product** of presheaves of $\mathcal{O}$ -modules

$$(\mathcal{F} \otimes_{p, \mathcal{O}} \mathcal{G})(U) := \mathcal{F}(U) \otimes_{\mathcal{O}(U)} \mathcal{G}(U)$$

Especially,

$$(\mathcal{O}_2 \otimes_{p, \mathcal{O}_1} \mathcal{G})(U) := \mathcal{O}_2(U) \otimes_{\mathcal{O}_1(U)} \mathcal{G}(U)$$

p for presheaf.

So we got a functor $PMod(\mathcal{O}_1) \rightarrow PMod(\mathcal{O}_2)$.

The construction is the categorical tensor product in $PMod(\mathcal{O})$:

$$\begin{array}{ccc} & \mathcal{O} & \\ \swarrow & & \searrow \\ \mathcal{F} & & \mathcal{G} \\ \searrow & & \swarrow \\ & \mathcal{F} \otimes_{p, \mathcal{O}} \mathcal{G} & \end{array}$$

♠ **Proposition 1.1.1.16** With $X, \mathcal{O}_1, \mathcal{O}_2, \mathcal{F}, \mathcal{G}$ as above, we have

$$Hom_{\mathcal{O}_1}(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}) = Hom_{\mathcal{O}_2}(\mathcal{O}_2 \otimes_{p, \mathcal{O}_1} \mathcal{G}, \mathcal{F})$$

Adjoint: Restriction $\dashv$ Change of Rings

◦ **Proof** Consider $\varphi \mapsto id_{\mathcal{O}_2} \otimes_{p, \mathcal{O}_1} \varphi : \mathcal{O}_2 \otimes_{p, \mathcal{O}_1} \mathcal{G} \rightarrow \mathcal{O}_2 \otimes_{p, \mathcal{O}_1} \mathcal{F}_{\mathcal{O}_1} = \mathcal{F}$
 The inverse: Restriction $\varphi \mapsto \varphi|1 \otimes_{p, \mathcal{O}_1} \mathcal{G}$

□

1.1.2 Sheaves

Definition 1.1.2.1 A **sheaf** is a presheaf $\mathcal{F}$ which satisfies: Let $U = \cup U_i$ be an open covering

- (Identity) Let $U = \cup U_i$ be an open covering. If

$$s|_{U_i} = t|_{U_i}$$

Then $s = t$;

- (Pasting) Given $s_i \in \mathcal{F}(U_i)$, such that

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$$

Then there exists some $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$.

Remark In the case of sheaves over Sets, the element $\mathcal{F}(\emptyset)$ can be deduced to be $\bullet$.

Remark Construction for Sets

$$\mathcal{F}(U \cup V) = \mathcal{F}(U) \times \mathcal{F}(V); U, V \in \text{Open}(X), U \cap V = \emptyset$$

♣ Example 1.1.2.2 ...

Definition 1.1.2.3 (Constant Sheaf) Let X be a topological space. A be a set

The **constant sheaf** with value A denoted $\underline{A}$ or $\underline{A}_X$ is the sheaf that assigns to an open $U \subseteq X$ the set of all locally constant maps $U \rightarrow A$:

$$\underline{A}_X(U) := \{f : U \rightarrow A | f \text{ is locally constant}\}$$

And restriction mappings given by restrictions of functions.

♣ Example 1.1.2.4 (Product Construction) Let $(A_x)_{x \in X}$ be a family of sets. Define:

$$\prod(U) := \prod_{x \in U} A_x$$

With following restriction maps:

$$(a_x)_{x \in U} \mapsto (a_x)_{x \in V}; U \subseteq V$$

And

- Identity: Omitted;
- Pasting: Well-defined.

♣ Example 1.1.2.5 For abelian groups M_x , construction

$$\mathcal{F} : U \mapsto \bigoplus_{x \in U} M_x$$

may be not a sheaf (but still a presheaf): Such $\text{supp}(s)$; $s \in \bigoplus_{x \in X} M_x$ must be finite-supported.

Furthermore:

♠ Proposition 1.1.2.6 For such $\mathcal{F}$: Every open of X is quasi-compact, then $\mathcal{F}$ is a sheaf.

◦ **Proof** Pasting: For a family of $s_i \in \mathcal{F}(U_i)$, then select a finite covering, and exists $s|_{U_i} = s_i, s \in M$ with finite number of supports. $\square$

Abelian Sheaves

Definition 1.1.2.7

- An **abelian sheaf** on X is an abelian presheaf on X such that: The underlying presheaf is a sheaf;
- The category of sheaves of abelian groups is denoted $Ab(X)$.

The conditions is equivalent to the exactness of

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \longrightarrow \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

With maps $s \mapsto \prod s|_{U_i}, (s_i)_{i \in I} \mapsto (s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j})_{i,j}$.

Sheaves of Algebraic Structures

Definition 1.1.2.8 In general: Let X be a topological space, $\mathcal{C}$ be a sheaf with product.

A presheaf with value in $\mathcal{C}$ is a sheaf if for every open covering, the diagram

$$0 \longrightarrow \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \cap U_j)$$

is an equalizer diagram in the category $\mathcal{C}$.

Now, with the faithful functor $F : \mathcal{C} \rightarrow \mathbf{Sets}$

$$0 \longrightarrow F(\mathcal{F}(U)) \longrightarrow F(\prod_i \mathcal{F}(U_i)) \rightrightarrows F(\prod_{i,j} \mathcal{F}(U_i \cap U_j))$$

F commutes with equalizer $\Leftrightarrow \mathcal{C}$ has limits and F commutes with limits.

F is needed to reflect isomorphisms: Bijection $\Leftrightarrow$ Isomorphism

◇ **Lemma 1.1.2.9** Suppose the category $\mathcal{C}$ and the functor $F : \mathcal{C} \rightarrow \mathbf{Sets}$ has the following properties:

- F is faithful;
- $\mathcal{C}$ has limits and F commutes with them;
- The functor F reflects isomorphisms.

Let X be a topological space, sheaf $\mathcal{F}$ on X is a presheaf with values in $\mathcal{C}$. Then, $\mathcal{F}$ is a sheaf iff the underlying presheaf is a sheaf.

◦ **Proof**

- $\Rightarrow$ Assume $\mathcal{F}$ is a sheaf: We see that $F(\mathcal{F}(U))$ is a equalizer of teh corresponding digram of sets. Hence $F(\mathcal{F})$ is a sheaf of sets.
- $\Leftarrow$ If $F(\mathcal{F})$ is a sheaf, then let $E \in \text{Ob}(\mathcal{C})$ be the equalizer of two parallel arrows: Consider a covering $\{U_i\}$ of U , then

$$E \longrightarrow \prod \mathcal{F}(U_i) \rightrightarrows \mathcal{F}(U_1 \cap U_j)$$

The equalizer exists, because of the existence of limits. So, there exists a canonical morphism

$$F(U) \rightarrow E$$

By the universal property of equalizer. Apply functor F , the induced map

$$F(F(U)) \rightarrow F(E)$$

is an isomorphism: Notice that, the equalizer commutes with functor F .

□

This lemma can be applied on:

- Groups;
- Rings;
- Algebras over a fixed Ring;
- Modules over a fixed Ring;
- Vector Space over a fixed Field.

Sheaves of Modules

Definition 1.1.2.10 (Sheaves of Modules)

- A sheaf of $\mathcal{O}$ -modules is a presheaf of $\mathcal{O}$ -modules $\mathcal{F}$, such that the underlying presheaf of abelian groups $\mathcal{F}$ is a sheaf;
- A **morphism** of sheaves of $\mathcal{O}$ -modules is a morphism of presheaves of $\mathcal{O}$ -modules;
- Given sheaves of $\mathcal{O}$ -modules $\mathcal{F}, \mathcal{G}$. Then denote

$$\text{Hom}_{\mathcal{O}}(\mathcal{F}, \mathcal{G})$$

the set of morphisms of sheaves of $\mathcal{O}$ -modules;

- The category of sheaves of $\mathcal{O}$ -modules is denoted $\text{Mod}(\mathcal{O})$.

1.1.3 Stalks

Definition 1.1.3.1 Let $\mathcal{F}$ be a sheaf over sets. Define: The **stalk of $\mathcal{F}$** at $x \in X$ is the set

$$\mathcal{F}_x := \text{colim}_{x \in U} \mathcal{F}(U)$$

It has a directed system:

$$(\mathcal{N}(x), \leq)$$

Where

- $\mathcal{N}(x) := \{U | x \in U, U \in \text{Open}(X)\}$;
- $U \geq U' \Leftrightarrow U \subseteq U'$;
- Transition Maps in the system are given by the restriction maps;

Equivalently,

$$\mathcal{F}_x = \mathcal{N}(x) / \sim$$

Definition 1.1.3.2 There exists canonical map

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x \quad s \mapsto \prod_{x \in U} (U, s)$$

Separated Property

◇ **Lemma 1.1.3.3** Let $\mathcal{F}$ be a sheaf of sets on the topological space X . For every open $U \subseteq X$, the map

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x$$

is injective.

◦ **Proof** Now suppose $s, s' \in \mathcal{F}(U)$ with same image, then

$$s|_{V^x} = s'|_{V^x}; U = \bigcup_{x \in U} V^x$$

By uniqueness of sheaf, $s = s'$. □

Definition 1.1.3.4 Let X be a topological space, and $\mathcal{F}$ be a presheaf on X over sets. Then, $\mathcal{F}$ is separated if for every open $U \subseteq X$,

$$\mathcal{F}(U) \rightarrow \prod_{x \in U} \mathcal{F}_x$$

is injective.

Functoriality

♠ **Proposition 1.1.3.5** $\mathcal{F} \mapsto \mathcal{F}_x$ is functorial in the presheaf $\mathcal{F}$. It gives a functor

$$PSh(X) \rightarrow Sets \quad \mathcal{F} \mapsto \mathcal{F}_x$$

◦ **Proof** Consider the following diagram:

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \mathcal{F}(U) & \longrightarrow & \mathcal{F}_x \\ \downarrow \varphi & & \downarrow \varphi(U) & & \downarrow \varphi_x \\ \mathcal{G} & \longrightarrow & \mathcal{G}(U) & \longrightarrow & \mathcal{G}_x \end{array}$$

And $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ given by $[(U, s)] \mapsto [(U, \varphi(s))]$. □

♣ Example 1.1.3.6

- Constant Presheaf:
- Germ:
- Consider a family of sets $\{A_x\}_{x \in X}$. Then, consider the sheaf

$$\mathcal{F} : U \mapsto \prod_{x \in U} A_x$$

Then, we have the special case, that x is in general not equal to the st A_x . For example:

Stalks of Abelian Presheaves

◇ **Lemma 1.1.3.7** Let X be a topological space. Let $\mathcal{F}$ be a presheaf of abelian groups on X . There exists a unique structure of an abelian group on $\mathcal{F}_x$ such that for every $U \subseteq X$ open, $x \in U$ the map

$$\mathcal{F}(U) \rightarrow \mathcal{F}_x$$

is a group homomorphism. Moreover:

$$\mathcal{F}_x = \operatorname{colim}_{x \in U} \mathcal{F}(U)$$

holds in category $\mathcal{A}b$.

◦ **Proof** Define the addition on $\mathcal{F}_x$ by

$$[(s, U)] + [(t, V)] := [(s|_{U \cap V} + t|_{U \cap V}, U \cap V)]$$

□

Stalks of Presheaves of Algebraic Structures

◇ **Lemma 1.1.3.8** Let $\mathcal{C}$ be a category, $F : \mathcal{C} \rightarrow \mathbf{Sets}$ be a functor. Assume that:

- F is faithful;
- Directed colimits exist in $\mathcal{C}$ and F commutes with them;

Let X be a topological space, $x \in X$. Let $\mathcal{F}$ be a presheaf with values in $\mathcal{C}$. Then:

$$\mathcal{F}_x = \operatorname{colim}_{x \in U} \mathcal{F}(U)$$

exists in $\mathcal{C}$. Its underlying set is equal to the stalk of underlying presheaf of sets of $\mathcal{F}$.

Furthermore, the construction $\mathcal{F} \mapsto \mathcal{F}_x$ is a functor from the category of presheaves with values in $\mathcal{C}$ to $\mathcal{C}$.

◦ **Proof** Notice that F commutes with directed colimits.

□

Inparticular:

- Groups;
- Rings;
- Modules over fixed ring;
- Vector Spaces over a fixed Field;

By definition,

$$\mathcal{F}(U) \rightarrow \mathcal{F}_x$$

are morphisms in the category $\mathcal{C}$ which turn into the corresponding maps for the underlying sheaf of sets. Thus,

$$\mathcal{F}(U) \rightarrow \mathcal{F}_x$$

Stalks of Presheaves of Modules

◇ **Lemma 1.1.3.9** Let X be a topological space, $\mathcal{O}$ a presheaf of rings on X .
Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$ -modules, $x \in X$. The canonical map

$$\mathcal{O}_x \times \mathcal{F}_x \rightarrow \mathcal{F}_x$$

coming from the multiplication map $\mathcal{O} \times \mathcal{F} \rightarrow \mathcal{F}$ defines a $\mathcal{O}_x$ -module structure on the abelian group $\mathcal{F}_x$.

◦ **Proof ...** □

◇ **Lemma 1.1.3.10** (Tensor Product of Stalks) Let X be a topological space, and $\mathcal{O} \rightarrow \mathcal{O}'$ be a morphism of presheaves of rings on X .

Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$ -modules, $x \in X$, we have

$$\mathcal{F}_x \otimes_{\mathcal{O}_x} \mathcal{O}'_x = (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{O}')_x$$

as $\mathcal{O}'_x$ -module.

◦ **Proof ...** □

Algebraic Structure

Definition 1.1.3.11 A type of algebraic structure is given by a category $\mathcal{C}$ and a functor $F : \mathcal{C} \rightarrow \mathbf{Sets}$ with following properties:

- F is faithful;
- $\mathcal{C}$ has limits and F commutes with limits;
- $\mathcal{C}$ has filtered colimits and F commutes with them;
- F reflects isomorphism.

◇ **Lemma 1.1.3.12** The following categories, endowed with the obvious forgetful functor, defines types of algebraic structures:

- Category of Pointed Sets;
- Category of Abelian Groups;
- Category of Groups;
- Category of Monoids;
- Category of Rings;
- Category of R -modules for a fixed ring R ;
- Category of Lie Algebras over Field.

◦ **Proof Omitted.** □

◇ **Lemma 1.1.3.13** Let $(\mathcal{C}, F)$ be a type of algebraic structure:

- $\mathcal{C}$ has a final object 0 , and $F(0) = \{*\}$;

- C has products and $F(\prod A_i) = \prod F(A_i)$;
- C has fibre product, and

$$F(A \times_B C) = F(A) \times_{F(B)} F(C)$$

- C has equalizers, if $E \rightarrow A$ is the equalizer of $a, b : A \rightarrow B$, then

$$F(E) \rightarrow F(A)$$

is the equalizer of $F(a), F(b) : F(A) \rightarrow F(B)$.

- $A \rightarrow B$ is a monomorphism iff $F(A) \rightarrow F(B)$ is injective.
- $A \rightarrow B$ is an epimorphism iff $F(A) \rightarrow F(B)$ is surjective.
- Given $A_1 \rightarrow A_2 \rightarrow \dots$, then colimit A_i exists, and

$$F(\text{colim} A_i) = \text{colim}(F(A_i))$$

•

◦ **Proof** Omitted. Check definition. □

◇ **Lemma 1.1.3.14** Let (C, F) be a type of algebraic structure. Suppose that:

$$A, B, C \in \text{Ob}(C)$$

Let $f : A \rightarrow B$ and $g : C \rightarrow B$ be morphisms of C . If $F(g)$ is injective, and $\text{Im}(F(f)) \subseteq \text{Im}(F(g))$, then f factors as

$$f = g \circ t, t : A \rightarrow C$$

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow & \uparrow g \\ & & C \end{array}$$

◦ **Proof** Consider $A \times_B C$:

$$\begin{array}{ccc} A \times_B C & \longrightarrow & C \\ \downarrow & & \downarrow \\ A & \longrightarrow & B \end{array}$$

Then, $A \times_B C \rightarrow A$ is bijective: Use the property 1.1.3.13, F commutes with fibre product,

- Surjectivity: By condition.
- Injectivity: Consider the set-theoretic mapping

$$F(A \times_B C) \rightarrow A \quad (a, c) \mapsto a$$

Notice that $F(g)$ is injective, thus there exists a unique $c \in F(C)$ such that $F(f)(a) = F(g)(c)$, pair (a, c) is unique. □

★ **Corollary 1.1.3.15** Apply the lemma on the following diagram:

$$\begin{array}{ccc} A & \longrightarrow & B \\ & & \downarrow \\ C & \longrightarrow & D \end{array}$$

Suppose $C \rightarrow D$ is injective, $\text{Im}(A \rightarrow B \rightarrow D) \subseteq \text{Im}(C \rightarrow D)$, then we have:

$$\begin{array}{ccc} A & \longrightarrow & B \\ \downarrow & & \downarrow \\ C & \longrightarrow & D \end{array}$$

Germes

Definition 1.1.3.16 The germs are defined as following: For $s \in \mathcal{F}(U), t \in \mathcal{F}(V)$ we have

$$(U, s) \sim (V, t) \Leftrightarrow s|_{U \cap V} = t|_{U \cap V}$$

Then:

♠ **Proposition 1.1.3.17** (Germes-Stalks) The following classes are the same

$$\mathcal{F}_x = \{(U, s) | U \in \text{Open}(X), s \in \mathcal{F}(U)\} / \sim$$

◦ **Proof** We will verify that this set satisfies the universal property: If there exists M and $f_U : \mathcal{F}(U) \rightarrow M$, then $h : \mathcal{F}_x \rightarrow M$ exists, via

$$h([U, s]) = f_U(s)$$

Then, proved. □

1.1.4 Exactness

Monomorphism and Epimorphism of Morphism

♠ **Proposition 1.1.4.1** (The Monomorphism and Epimorphism of Section) For $\varphi : \mathcal{F} \rightarrow \mathcal{G}$:

- φ is mono $\Leftrightarrow$ For all open $U \subseteq X, \varphi(U)$ is mono;
- $\varphi(U)$ is epi for any $U \Rightarrow \varphi$ is epi. Conversely not hold.

◦ **Proof**

- Easy to check,
- Omitted. And if all are epi, then it is **flasque**

□

Exactness

◇ **Lemma 1.1.4.2** Let X be a topological space, $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves of sets on X :

- The map φ is a monomorphism in the category of sheaves iff for all $x \in X$, the map $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is injective;
- The map φ is an epimorphism in the category of sheaves iff for all $x \in X$, the map $\varphi_x : \mathcal{F}_x \rightarrow \mathcal{G}_x$ is surjective;
- Isomorphism $\Leftrightarrow$ bijective;

◦ **Proof**

- $\Rightarrow$ Suppose $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is monomorphism. Then, consider φ_x , with open neighborhood $U \ni x$, then

$$\varphi_x([a]) = \varphi_x([b]) \Rightarrow$$

And stalk is a filtered colimit, commutes with morphism φ . Thus,

$$\Leftrightarrow (\varphi(U)(a)) = (\varphi(U)(b))$$

- $\Leftarrow$ If φ_x is mono, then φ According to the commutativity of colimit

$$\varphi(U)(s) = \varphi(U)(t) \Rightarrow (\varphi(U)(s))_x = (\varphi(U)(t))_x, x \in U$$

Now, select converging of U . Thus $s = t$.

- Similarly
- Omitted.

□

Definition 1.1.4.3 Let X be a topological space

- A presheaf $\mathcal{F}$ is called a **subpresheaf** of a presheaf $\mathcal{G}$ if $\mathcal{F}(U) \subseteq \mathcal{G}(U)$ for all open $U \subseteq X$ such that the restriction maps of $\mathcal{G}$ induce the restriction map of $\mathcal{F}$.

If $\mathcal{F}, \mathcal{G}$ are sheaves, then $\mathcal{F}$ is called a **subsheaf** of $\mathcal{G}$. We sometimes indicate this by the notation

$$\mathcal{F} \subseteq \mathcal{G}$$

- A morphism of presheaves of sets $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is called **injective** iff $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective, $U \subseteq X$ opens.
- A morphism of presheaves of sets $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is called **surjective** iff $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is surjective.
- Sheaves of Sets injective iff $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective;
- A morphism of sheaves of sets $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ on X is called **surjective** iff for every open U of X , and every section $s \in \mathcal{G}(U)$, there exists an open covering $U = \cup U_i$ such that

$$s|_{U_i} \in \text{Im}(\mathcal{F}(U_i) \rightarrow \mathcal{G}(U_i))$$

◇ **Lemma 1.1.4.4** Let X be a topological space:

- Epimorphisms/Monomorphisms in the category of presheaves are exactly the surjective/injective maps of presheaves;
- Epimorphisms/Monomorphisms in the category of sheaves are exactly the maps which are surjective/injective on all stalks.

◦ **Proof** Omitted.

□

Algebraic Construction

◇ **Lemma 1.1.4.5** Let X be a topological space, (C, F) be a type of algebraic structure.

Suppose $\mathcal{F}, \mathcal{G}$ are sheaves on X with values in C , $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be the map of underlying sheaves of sets.

If for all $x \in X$, the map $\mathcal{F}_x \rightarrow \mathcal{G}_x$ is a morphisms of algebraic structures, then φ is a morphism of sheaves of algebraic structures.

◦ **Proof** Consider the following diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \longrightarrow & \prod_{x \in U} \mathcal{F}_x \\ \downarrow & & \downarrow \\ \mathcal{G}(U) & \longrightarrow & \prod_{x \in U} \mathcal{G}_x \end{array}$$

And except $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$. By the lemma 1.1.3.15, $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is also morphism of algebraic structure, thus we conclude that $\mathcal{F} \rightarrow \mathcal{G}$ is algebraic. $\square$

1.1.5 Sheafifications

Definition 1.1.5.1 Let $\mathcal{F}$ be a presheaf. For every open $U \subseteq X$, we define

$$\mathcal{F}^\#(U) := \{(s_u)_{u \in U} \in \prod_{u \in U} \mathcal{F}_u \mid *\}$$

where $*$ is the property: For every $u \in U$, there exists a neighborhood V with $\sigma \in \mathcal{F}(V)$ such that

$$s_v = (\sigma, V) \in \mathcal{F}_v; v \in V$$

with restriction maps (projection), for $V \subseteq U$,

$$\mathcal{F}^\#(U) \rightarrow \mathcal{F}^\#(V) \quad \prod_{u \in U} \mathcal{F}_u \rightarrow \prod_{v \in V} \mathcal{F}_v$$

Define sheaf $\prod \mathcal{F} : U \mapsto \prod_{u \in U} \mathcal{F}_u$ There exists morphism of presheaves

$$\mathcal{F} \rightarrow \mathcal{F}^\# \rightarrow \prod \mathcal{F}$$

with following diagrams:

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}^\#(U) & \longrightarrow & \prod_{u \in U} \mathcal{F}_u \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}(V) & \longrightarrow & \mathcal{F}^\#(V) & \longrightarrow & \prod_{v \in V} \mathcal{F}_v \end{array}$$

♠ **Proposition 1.1.5.2** The presheaf $\mathcal{F}^\#$ is a sheaf.

◦ **Proof**

- Pasting: Suppose that there exists covering $U = \cup V_i$ with (V_i, s_i) such that

$$s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$$

Then, equivalently, if we write

$$\begin{aligned} s_i &= \prod_{v \in V_i} (s_v), s_j = \prod_{v \in V_j} (s'_v) \\ \Rightarrow s_v &= s'_v, v \in V_i \cap V_j \end{aligned}$$

Then, there exists construction

$$s = \prod_{v \in U} (s_v)$$

- Identity: Omitted.

□

Stalks

◇ **Lemma 1.1.5.3** Let X be a topological space. Let $\mathcal{F}$ be a presheaf of sets on X . Let $x \in X$, then

$$\mathcal{F}_x = \mathcal{F}_x^\#$$

- **Proof** Notice that:

$$\text{colim}_{x \in U} \left(\prod_{v \in V} (s_v) \right) = s_x$$

□

◇ **Lemma 1.1.5.4** Let $\mathcal{F}$ be a sheaf, then $\mathcal{F}^\$ = \mathcal{F}$.

- **Proof** ...

□

◇ **Lemma 1.1.5.5** Let $\mathcal{F}$ be a presheaf of sets on X . Any map $\mathcal{F} \rightarrow \mathcal{G}$ into a sheaf of sets factors uniquely as

$$\mathcal{F} \rightarrow \mathcal{F}^\# \rightarrow \mathcal{G}$$

- **Proof** Consider the following diagram

$$\begin{array}{ccccc} \mathcal{F} & \longrightarrow & \mathcal{F}^\# & \longrightarrow & \prod(\mathcal{F}) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}^\# & \longrightarrow & \prod(\mathcal{G}) \end{array}$$

□

This implies, that there is an adjoint pair of functors:

$$i : Sh(X) \rightarrow Psh(X) \quad \# : Psh(X) \rightarrow Sh(X)$$

♠ **Proposition 1.1.5.6** There is an adjoint pair of functors $i : Sh(X) \rightarrow Psh(X)$ and $\# : Psh(X) \rightarrow Sh(X)$, the formula

$$Mor_{Psh(X)}(\mathcal{F}, i(\mathcal{G})) = Mor_{Sh(X)}(\mathcal{F}^\#, \mathcal{G})$$

holds for sheaf $\mathcal{G}$, presheaf $\mathcal{F}$.

- **Proof** Consider the diagram above.

□

◇ **Lemma 1.1.5.7** Let X be a topological space. A presheaf $\mathcal{F}$ is separated iff the canonical map $\mathcal{F} \rightarrow \mathcal{F}^\#$ is injective.

◦ **Proof** By definition of Construction. □

◇ **Lemma 1.1.5.8** Let X be a topological space. The **sheafification** of a surjective/injective morphism of presheaves of sets is surjective/injective.

◦ **Proof** According to our discussion of monomorphism/epimorphism of stalk. □

Sheafification of Abelian Presheaves

◇ **Lemma 1.1.5.9** We have: For (pre)sheaf of abelian group we have

$$\mathcal{F}_x = \mathcal{F}_x^\#$$

◦ **Proof** Category $\mathcal{A}b$ is also algebraic, preverse filtered colimit. □

◇ **Lemma 1.1.5.10** Let X be a topological space, $\mathcal{F}$ be a presheaf of sets on X . Let $U \subseteq X$ open, there exists a canonical fibre product diagram:

$$\begin{array}{ccc} \mathcal{F}^\#(U) & \longrightarrow & (\prod \mathcal{F})(U) \\ \downarrow & & \downarrow \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \prod(\mathcal{F})_x \end{array}$$

where the maps are the following:

- The left vertical map has components

$$\mathcal{F}^\#(U) \rightarrow \mathcal{F}_x^\# = \mathcal{F}_x$$

- The top horizontal map comes from the map of presheaves $\mathcal{F} \rightarrow \prod(\mathcal{F})$;
- The right vertical map has obvious component maps $\prod \mathcal{F}(U) \rightarrow \prod_{x \in U} (\prod \mathcal{F})_x$;
- The bottom horizontal map has components $\mathcal{F}_x \rightarrow (\prod \mathcal{F})_x$ which come from the map of presheaves $\mathcal{F} \rightarrow \prod(\mathcal{F})$.

◦ **Proof**

- By the conclusion above.
- Notice: $\prod(\mathcal{F})$ is a sheaf, so there exists factorization

$$\mathcal{F} \rightarrow \mathcal{F}^\# \rightarrow \prod(\mathcal{F})$$

- By the morphism of $\prod \mathcal{F}$;
- Obvious. By taking $\mathcal{F}_x \rightarrow (\prod \mathcal{F})_x$ because of category $\mathcal{A}b$.

□

◇ **Lemma 1.1.5.11** Let X be a topological space, $\mathcal{F}$ be an abelian presheaf on X . Then, there exists a unique structure of abelian sheaf on $\mathcal{F}^\#$ such that:

$$\mathcal{F} \rightarrow \mathcal{F}^\#$$

is a morphism of abelian presheaves.

Moreover, the following adjointness property holds:

$$\text{Mor}_{\text{Ab}(X)}(\mathcal{F}, i(\mathcal{G})) = \text{Mor}_{b(X)}(\mathcal{F}^\#, \mathcal{G})$$

◦ **Proof** Firstly, let $\mathcal{F}$ be an abelian sheaf, then $\mathcal{F}_x$ is abelian. Consider the morphism of abelian group

$$\mathcal{F} \rightarrow (\prod \mathcal{F})$$

$\mathcal{F}^\#(U)$ is a subgroup of $(\prod \mathcal{F})(U)$ by definition. Especially, as above, $\mathcal{F}^\#(U)$ is a fibre product.

$\mathcal{F}_x \rightarrow \mathcal{F}_x^\#$ is an isomorphism of abelian group, to prove the adjointness: We use the adjointness property of sets. □

Sheafification of Presheaves of Algebraic Structures

◇ **Lemma 1.1.5.12** Let X be a topological space. Let (C, F) be a type of algebraic structure. Let $\mathcal{F}$ be a presheaf with values in C on X . Then there exists a sheaf $\mathcal{F}^\#$ with values in C , and a morphism

$$\mathcal{F} \rightarrow \mathcal{F}^\#$$

of presheaves with values in C with the following properties:

- The map $\mathcal{F} \rightarrow \mathcal{F}^\#$ identifies the underlying presheaf of sets of $\mathcal{F}$;
- For any morphism $\mathcal{F} \rightarrow \mathcal{G}$, where $\mathcal{G}$ is a sheaf with values in C there exists a unique factorization

$$\mathcal{F} \rightarrow \mathcal{F}^\# \rightarrow \mathcal{G}$$

◦ **Proof** Same as proof above. Define $\mathcal{F}^\#(U)$ to be the fibre product in C of the diagram

$$\begin{array}{ccc} & \prod(\mathcal{F})(U) & \\ & \downarrow & \\ \prod_{x \in U} \mathcal{F}_x & \longrightarrow & \prod_{x \in U} \mathcal{F}_x^\# \end{array}$$

□

Sheafification of Presheaves of Modules

◇ **Lemma 1.1.5.13** Let X be a topological space, $\mathcal{O}$ presheaf of rings on X . Let $\mathcal{F}$ be a presheaf of $\mathcal{O}$ -modules, $\mathcal{O}^\#$ be the sheafification and $\mathcal{F}^\#$ also be the sheafification, then there exists module structure of sheafification

$$\mathcal{O}^\# \times \mathcal{F}^\# \rightarrow \mathcal{F}^\#$$

which makes the diagram

$$\mathcal{O} \times \mathcal{F} \quad \mathcal{F}$$

$$\mathcal{O}^\# \times \mathcal{F}^\# \quad \mathcal{F}^\#$$

Moreover, $\mathcal{F} \rightarrow \mathcal{G}$ factors uniquely through $\mathcal{F} \rightarrow \mathcal{F}^\# \rightarrow \mathcal{G}$ is a morphism of $\mathcal{O}^\#$ -modules.

◦ **Proof ...**

□

★ **Corollary 1.1.5.14** Formula

$$\text{Mor}_{\text{PMod}(\mathcal{O})}(\mathcal{F}, i\mathcal{G}) = \text{Mor}_{\text{Mod}(\mathcal{O})}(\mathcal{F}^\#, \mathcal{G})$$

◦ **Proof ...**

□

Definition 1.1.5.15 Define:

- Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_2$ -module, with module homomorphism

$$\rho : \mathcal{O}_1 \rightarrow \mathcal{O}_2$$

Then: $\mathcal{F}$ has a $\mathcal{O}_1$ -module structure, with

$$\mathcal{O}_1 \times \mathcal{F} \rightarrow \mathcal{F} \quad (s, t) \mapsto \rho(U)(s) \cdot t$$

Define the following module as $\mathcal{F}_{\mathcal{O}_1}$.

- Tensor Product: For $\mathcal{O}_1 \rightarrow \mathcal{O}_2$, define

$$\mathcal{O}_2 \otimes_{\mathcal{O}_1} \mathcal{F} := (\mathcal{O}_2(U) \otimes_{\mathcal{O}_1(U)} \mathcal{F}(U))^\#$$

◇ **Lemma 1.1.5.16** (Adjointness of Modules) With $X, \mathcal{O}_1, \mathcal{O}_2$ and $\mathcal{F}, \mathcal{G}$ as above there exists a canonical bijection

$$\text{Hom}_{\mathcal{O}_1}(\mathcal{G}, \mathcal{F}_{\mathcal{O}_1}) = \text{Hom}_{\mathcal{O}_2}(\mathcal{O}_2 \otimes_{\mathcal{O}_1} \mathcal{G}, \mathcal{F})$$

In other words, the restriction and change of rings are adjoint to each other.

◦ **Proof ...**

□

★ **Corollary 1.1.5.17** Let X be a topological space, $\mathcal{O} \rightarrow \mathcal{O}'$ be a morphism of sheaves of rings on X , $\mathcal{F}$ be a sheaf of $\mathcal{O}$ -module. We have

$$\mathcal{F}_x \otimes_{\mathcal{O}_x} \mathcal{O}'_x = (\mathcal{F} \otimes_{\mathcal{O}} \mathcal{O}')_x$$

◦ **Proof ...**

□

1.1.6 Limits and Colimits

Presheaves

◇ **Lemma 1.1.6.1** Let X be a topological space, $\mathcal{I} \rightarrow Psh(X), i \mapsto \mathcal{F}_i$ be a diagram:

- Both $\lim_i \mathcal{F}_i$ and $\operatorname{colim}_i \mathcal{F}_i$ exists;
- For any $U \subseteq X$ open we have

$$\begin{cases} \lim_i \mathcal{F}_i(U) = (\lim_i \mathcal{F}_i)(U) \\ \operatorname{colim}_i \mathcal{F}_i(U) = (\operatorname{colim}_i \mathcal{F}_i)(U) \end{cases}$$

- Let $x \in X$ be point, then in general

$$(\lim_i \mathcal{F}_i)_x \neq \lim_i (\mathcal{F}_i)_x$$

If index category $\mathcal{I}$ is finite, then equal.

- Let $x \in X$, we always have

$$(\operatorname{colim}_i \mathcal{F}_i)_x = \operatorname{colim}_i \mathcal{F}_{i,x}$$

◦ Proof

-
-
-
-

□

Sheaves

◇ **Lemma 1.1.6.2** Let X be a topological space, $\mathcal{I} \rightarrow Psh(X), i \mapsto \mathcal{F}_i$ be a diagram:

- $\lim_i \mathcal{F}_i$ and $\operatorname{colim}_i \mathcal{F}_i$ exist;
- For any open $U \subseteq X$, we have

$$\begin{aligned} (\lim_i \mathcal{F}_i)(U) &= \lim_i (\mathcal{F}_i(U)) \\ \operatorname{colim}_i \mathcal{F}_i &= (U \mapsto \operatorname{colim}_i \mathcal{F}_i(U))^{\#} \end{aligned}$$

- Let $x \in X$ be a point. In general the stalk of $\lim_i \mathcal{F}_i$ at x is not equal to the limit of stalks.
But: If the index category is finite, then the two operations commute.
- Let $x \in X$, we always have

$$(\operatorname{colim}_i \mathcal{F}_i)_x = \operatorname{colim}_i \mathcal{F}_{i,x}$$

- Sheafification $(-)^{\#} : Psh(X) \rightarrow Sh(X)$ commutes with all colimits, finite limits. But it does not commute with all limits.

◦ **Proof**

-
-
-
-
-
-

□

◇ **Lemma 1.1.6.3** Let X be a topological space, I be a directed set. Let $(\mathcal{F}_i, \varphi_{ii'})$ be a system of sheaves of sets over I . Let $U \subseteq X$ be an open subset. Consider the canonical map

$$\Psi : \operatorname{colim}_i \mathcal{F}_i(U) \rightarrow (\operatorname{colim}_i \mathcal{F}_i)(U)$$

-
-
-
-

◦ **Proof ...**

□

♣ **Example 1.1.6.4** ...

1.2 Morphisms

Definition 1.2.0.1

- A morphism of presheaves is a family of mapping $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ with: For $U \subseteq V$,

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow & & \downarrow \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \end{array}$$

- The morphism of sheaves is the morphism of presheaves.
- (Restrictions) Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism in $\mathcal{Psh}(X)$. Then, the **restriction** on $U \subseteq X$ is defined as:

$$\varphi|_U(V) : \mathcal{F}(U \cap V) \rightarrow \mathcal{G}(U \cap V) \quad s \mapsto \varphi(U \cap V)(s)$$

◇ **Lemma 1.2.0.2** ...◦ **Proof ...**

□

1.2.1 Continuous Maps and Sheaves

Let $f : X \rightarrow Y$ be a continuous map of topological spaces. We define the **pushforward** and **pullback** functors for presheaves and sheaves.

Pushforward

Definition 1.2.1.1

Let $\mathcal{F}$ be a presheaf of sets on X . Define the **pushforward**

$$f_* \mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$$

for open $V \subseteq Y$. With following restriction maps: For $V_1 \subseteq V_2 \subseteq Y$,

$$\begin{array}{ccc} f_* \mathcal{F}(V_2) & \xrightarrow{=} & \mathcal{F}(f^{-1}(V_2)) \\ \downarrow & & \downarrow \text{restriction for } \mathcal{F} \\ f_* \mathcal{F}(V_1) & \xrightarrow{=} & \mathcal{F}(f^{-1}(V_1)) \end{array}$$

This gives rise to a functor

...

◇ **Lemma 1.2.1.2** Let $f : X \rightarrow Y$ be a continuous map, $\mathcal{F}$ be a sheaf of sets on X . Then $f_* \mathcal{F}$ is a sheaf on Y .

◦ Proof

- Identity: For $U = \cup U_i, s = 0 \in f_* \mathcal{F}(U) \Leftrightarrow s = 0 \in \mathcal{F}(f^{-1}(U)) \Leftrightarrow s = 0$
- Pasting: $U = \cup U_i, s_i|_{U_i} = s_j|_{U_j} \Leftrightarrow s_i|_{f^{-1}(U_i)} = s_j|_{f^{-1}(U_j)} \Rightarrow \text{Exists } s \in \mathcal{F}(f^{-1}(U)), s|_{f^{-1}(U_i)} = s_i.$

□

★ **Corollary 1.2.1.3** The functor $f_* : PSh(X) \rightarrow PSh(Y)$ has a restriction on $Sh(X)$

$$f_* : Sh(X) \rightarrow Sh(Y)$$

◦ **Proof** Apply the decomposition $V = \cup V_j \subseteq Y \Rightarrow f^{-1}(V) = \cup f^{-1}(V_j) \subseteq X$ on identity lemma and pasting lemma. □

◇ **Lemma 1.2.1.4** Let $f : X \rightarrow Y$ and $Y \rightarrow Z$ be continuous maps of topological spaces. The functors

$$(g \circ f)_* = g_* \circ f_*$$

are equal.

◦ Proof ...

□

Definition 1.2.1.5 Via Yoneda's lemma, define left adjoint of the pushforward

$$Mor_{Psh(X)}(f_p \mathcal{G}, F) = Mor_{Psh(Y)}(\mathcal{G}, f_* F)$$

is uniquely determined.

Existence/Construction:

◇ **Lemma 1.2.1.6** Let $f : X \rightarrow Y$ be a continuous map. There exists a functor

$$f_p : Psh(Y) \rightarrow Psh(X) \quad (1.2)$$

which is left adjoint to f_* . For a presheaf $\mathcal{G}$ it is determined by the rule

$$f_p \mathcal{G}(U) = colim_{f(U) \subseteq V} \mathcal{G}(V)$$

◦ **Proof** $f_p \mathcal{G}$ is a presheaf.

- Objects: Given by set-theoretic colimit;
- Restriction Maps: Now suppose $U_1 \subseteq U_2$, then

$$\{V | f(U_1) \subseteq V\} \supseteq \{V | f(U_2) \subseteq V\}$$

- Colimit is functorial in $\mathcal{G}$ Thus f_p is functorial.
- (Universal Property) A useful remark: Canonical map

$$\mathcal{G}(U) \rightarrow f_p \mathcal{G}(f^{-1}(U)) = colim_{f(f^{-1}(U)) \subseteq V} \mathcal{G}(V)$$

Because of $f(f^{-1}(U)) \subseteq U$, U is a member of system. Then select the open subsets(subsystem) contained in U . (The collections are directed)

$$\begin{array}{ccc} & \mathcal{G}(U) & \\ \swarrow & & \searrow \\ \mathcal{G}(V_1) & & \mathcal{G}(V_2) \\ \searrow & & \swarrow \\ & colim_{f(f^{-1}(U)) \subseteq V} \mathcal{G}(V) & \end{array}$$

And then, there exists canonical map $i_{\mathcal{G}} : \mathcal{G} \rightarrow f_* f_p \mathcal{G}$.

Now: Suppose $\psi : f_p \mathcal{F} \rightarrow \mathcal{F}$ be a map of presheaves. Then the corresponding map

$$f_* \psi \circ i_{\mathcal{G}} : \mathcal{G} \rightarrow f_* f_p \mathcal{G} \rightarrow f_* \mathcal{F}$$

Another remark is canonical map $c_{\mathcal{F}} : f_p f_* \mathcal{F} \rightarrow \mathcal{F}$. If $f(U) \subseteq V$ in Y there exists a map $(f(U) \subseteq V \Rightarrow U \subseteq f^{-1}(f(U)) \subseteq f^{-1}(V))$

$$f_* \mathcal{F}(V) = \mathcal{F}(f^{-1}(V)) \rightarrow \mathcal{F}(U)$$

Now take the colimit

$$f_p f_* \mathcal{F}(V) = colim_{f(U) \subseteq V} f_* \mathcal{F}(V) \rightarrow \mathcal{F}(U)$$

Similarly, for $\varphi : \mathcal{G} \rightarrow f_* \mathcal{F}$, consider $f_p \varphi : f_p \mathcal{G} \rightarrow f_p f_* \mathcal{F}$. Then we desire the map

$$c_{\mathcal{F}} \circ f_p \varphi : f_p \mathcal{G} \rightarrow \mathcal{F}$$

Thus, we have

$$Mor_{Sh(Y)}(\mathcal{G}, f_* \mathcal{F}) \cong Mor_{Sh(X)}(f_p \mathcal{G}, \mathcal{F})$$

□

◇ **Lemma 1.2.1.7** Let $f : X \rightarrow Y$ be a continuous map. Let $x \in X$, $\mathcal{G}$ be a presheaf of sets on Y . There is a canonical bijection of stalks

$$(f_p \mathcal{G})_x = \mathcal{G}_{f(x)}$$

◦ **Proof**

$$\begin{aligned} (f_p \mathcal{G})_x &= \text{colim}_{x \in U} f_p \mathcal{G}(U) \\ &= \text{colim}_{x \in U} \text{colim}_{f(U) \subseteq V} \mathcal{G}(V) \\ &= \text{colim}_{f(x) \in V} \mathcal{G}(V) \\ &= \mathcal{G}_{f(x)} \end{aligned}$$

□

Definition 1.2.1.8 Let $\mathcal{G}$ be a sheaf of sets on Y . The **pullback sheaf** $f^{-1}\mathcal{G}$ is

$$f^{-1}\mathcal{G} := (f_p \mathcal{G})^\#$$

◇ **Lemma 1.2.1.9** In the category of sheaves,

$$\text{Mor}_{Sh(X)}(f^{-1}\mathcal{G}, \mathcal{F}) = \text{Mor}_{Sh(Y)}(\mathcal{G}, f_*\mathcal{F})$$

◦ **Proof** We have:

$$\begin{aligned} \text{Mor}_{Sh(X)}(f^{-1}\mathcal{G}, \mathcal{F}) &= \text{Mor}_{Psh(X)}(f_p \mathcal{G}, \mathcal{F}) \\ &= \text{Mor}_{Psh(Y)}(\mathcal{G}, f_*\mathcal{F}) \\ &= \text{Mor}_{Sh(Y)}(\mathcal{G}, f_*\mathcal{F}) \end{aligned}$$

□

◇ **Lemma 1.2.1.10** ...

◦ **Proof** ...

□

f -Map

Definition 1.2.1.11 Let $f : X \rightarrow Y$ be a continuous map. Let $\mathcal{F}$ be a sheaf of sets on X and let $\mathcal{G}$ be a sheaf of sets on Y .

An f -map $\xi : \mathcal{G} \rightarrow \mathcal{F}$ is a collection of maps

$$\xi_V : \mathcal{G}(V) \rightarrow \mathcal{F}(f^{-1}V)$$

Indexed by open $V \subseteq Y$ such that: For $V' \subseteq V \subseteq Y$,

$$\begin{array}{ccc}
\mathcal{G}(V) & \xrightarrow{\xi_V} & \mathcal{F}(f^{-1}(V)) \\
\downarrow & & \downarrow \\
\mathcal{G}(V') & \xrightarrow{\xi_{V'}} & \mathcal{F}(f^{-1}(V'))
\end{array}$$

◇ **Lemma 1.2.1.12** (The Bijection between f -Maps) Let $f : X \rightarrow Y$ be continuous map. There are bijection between the four sets:

- Set of Maps $\mathcal{G} \rightarrow f_*\mathcal{F}$;
- Set of Maps $f^{-1}\mathcal{G} \rightarrow \mathcal{F}$;
- Set of Maps f -maps $\xi : \mathcal{G} \rightarrow \mathcal{F}$;
- Set of all collections of maps

$$\xi_{U,V} : \mathcal{G}(V) \rightarrow \mathcal{F}(U); f(U) \subseteq X, V \subseteq Y, f(U) \subseteq V$$

and compatible with those restriction maps.

Functorially in $\mathcal{F} \in Sh(X), \mathcal{G} \in Sh(Y)$

◦ **Proof**

- (1) $\Leftrightarrow$ (2) Omitted;
- (1) $\Leftrightarrow$ (3) Now, the morphisms

$$\mathcal{G}(U) \rightarrow \mathcal{F}(f^{-1}(V))$$

It is actually the f -map.

- (2) $\Leftrightarrow$ (4) $Hom(f^{-1}\mathcal{G}, \mathcal{F})$ is correspondnet to a family of morphisms $\mathcal{G}(V) \rightarrow f_p\mathcal{G}(V) \rightarrow \mathcal{F}$
Conversely, define the forgetting structure $\xi_V := \xi_{f^{-1}(V), V}$.

□

Definition 1.2.1.13 (Composition) Suppose $f : X \rightarrow Y, g : Y \rightarrow Z$ be continuous maps of topological spaces. And $\varphi : \mathcal{G} \rightarrow \mathcal{F}$ is f -map and $\psi : \mathcal{H} \rightarrow \mathcal{G}$ is g -map. The **composition** of φ, ψ is the $(g \circ f)$ -map $\varphi \circ \psi$ defined by the commutativity

$$\begin{array}{ccc}
\mathcal{H}(W) & \xrightarrow{\quad} & \mathcal{F}(f^{-1}g^{-1}W) \\
& \searrow & \swarrow \\
& \mathcal{G}(g^{-1}W) &
\end{array}$$

◇ **Lemma 1.2.1.14** Suppose $f : X \rightarrow Y, g : Y \rightarrow Z$ be continuous map of topological spaces. Suppose $\mathcal{F}$ is a sheaf on $X, \mathcal{G}$ is a sheaf on $Y, \mathcal{H}$ is a sheaf on Z .

Let $\varphi : \mathcal{G} \rightarrow \mathcal{F}$ be an f map and $\psi : \mathcal{H} \rightarrow \mathcal{G}$ be an g map. Let $x \in X$ be a point. Then, the map on stalk $(\varphi \circ \psi)_x : \mathcal{H}_{g(f(x))} \rightarrow \mathcal{F}_x$ is the composition

$$\mathcal{H}_{g(f(x))} \rightarrow \mathcal{G}_{f(x)} \rightarrow \mathcal{F}_x$$

◦ **Proof ...**

□

Continuous Maps and Abelian Sheaves

We have following functors:

- $f_* : PAb(X) \rightarrow PAb(Y)$;
- $f_* : Ab(X) \rightarrow Ab(Y)$;
- $f_p : PAb(Y) \rightarrow PAb(X)$;
- $f^{-1} : Ab(Y) \rightarrow Ab(X)$;

Continuous Maps and Sheaves of Algebraic Structures

Continuous Maps and Sheaves of Modules

1.2.2 Open Immersions and (Pre)sheaves

Review Let

$$j : U \rightarrow X$$

be the inclusion of an open subset U into X . We have defined the following functors on sheaves

$$j_* : Sh(U) \rightarrow Sh(X) \quad j^{-1} : Sh(X) \rightarrow Sh(U)$$

which are adjointed.

◇ **Lemma 1.2.2.1** Let X be a topological space, $j : U \rightarrow X$ be the inclusion of an open subset U of X :

-
-
-
-
-

◦ **Proof ...**

□

Definition 1.2.2.2

-
-
-
-

1.2.3 Closed Immersions and (Pre)sheaves

Review Let $i : Z \rightarrow X$ be the inclusion. Then we have defined the adjointed functors

$$i_* : Sh(Z) \rightarrow Sh(X) \quad i^{-1} : Sh(X) \rightarrow Sh(Z)$$

◇ **Lemma 1.2.3.1** ...

◦ **Proof** ...

□

1.3 Ringed Spaces and Modules

Definition 1.3.0.1

- A **ringed space** is a pair $(X, \mathcal{O}_X)$ consisting:
 - A topological space X ;
 - The sheaf of rings $\mathcal{O}_X$;
- A morphism of ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$;
- Composition: $(g, g^\#) \circ (f, f^\#) = (g \circ f, f^\# \circ g^\#)$;

1.3.1 Morphisms of Ringed Spaces and Modules

Definition 1.3.1.1 Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces:

-
-

◇ **Lemma 1.3.1.2** Let

◦ **Proof** ...

□

◇ **Lemma 1.3.1.3** ...

◦ **Proof** ...

□

◇ **Lemma 1.3.1.4** Let $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let $\mathcal{G}$ be a sheaf of $\mathcal{O}_X$ -modules, $x \in X$, we have

$$(f_* \mathcal{G})_x = \mathcal{G}_{f(x)} \otimes_{\mathcal{O}_{Y, f(x)}} \mathcal{O}_{X, x}$$

◦ **Proof** ...

□

1.4 Constructions

1.4.1 Skyscraper Sheaves and Stalks

Definition 1.4.1.1

- Let $x \in X$ be a point. Denote

$$i_x : \{x\} \rightarrow X$$

The inclusion map. Let A be a set, and think of A as a sheaf on one-point space $\{x\}$. We call $i_{x,*}A$ the **skyscraper sheaf** at x with value A :

$$\begin{aligned} A(x) &:= A \\ \Rightarrow i_{x,*}A(U) &:= A(i_x^{-1}(U)), U \subseteq X \end{aligned}$$

- If A is an abelian group, then we think of $i_{x,*}A$ as a sheaf of abelian groups on X ;
- If A is an algebraic structure, then we think of $i_{x,*}A$ as a sheaf of algebraic structures.
- If $(X, \mathcal{O}_X)$ is a ringed space, then we think of

$$i_x : \{x\} \rightarrow X$$

as a morphism of ringed spaces

$$(\{x\}, \mathcal{O}_{\{x\}}) \rightarrow (X, \mathcal{O}_X)$$

and: If A is a $\mathcal{O}_{\{x\}}$ -module, then we think of $i_{x,*}A$ as a sheaf of $\mathcal{O}_X$ -module.

- We say a sheaf of sets $\mathcal{F}$ is a **skyscraper sheaf** if there exists a point $x \in X$ and a set such that

$$\mathcal{F} \cong i_{x,*}A$$

Similarly for:

- Abelian group;
- Algebraic Structures;
- Modules;

◇ **Lemma 1.4.1.2** Let X be a topological space, $x \in X$ a point, A a set. For any point $x' \in X$ the stalk of the skyscraper sheaf $\text{st } x$ with value A at x' is:

$$(i_{x,*}A)_{x'} = \begin{cases} A & x' \in \{\bar{x}\} \\ \{*\} & x' \notin \{\bar{x}\} \end{cases}$$

◦ **Proof ...**

□

◇ **Lemma 1.4.1.3** Let X be a topological space, and let $x \in X$ a point. The functors

$$\mathcal{F} \mapsto \mathcal{F}_x \quad A \mapsto i_{x,*}A$$

are adjoint. In formula

$$\text{Mor}_{\text{Sets}}(\mathcal{F}_x, A) = \text{Mor}_{\text{Sh}(X)}(\mathcal{F}, i_{x,*}(A))$$

◦ **Proof ...**

□

1.4.2 Bases and Sheaves

Presheaves

Definition 1.4.2.1 Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X :

- A presheaf $\mathcal{F}$ of sets on $\mathcal{B}$ is a rule which assigns to each $U \in \mathcal{B}$ a set $\mathcal{F}(U)$ and to each inclusion $V \subseteq U$ of elements in $\mathcal{B}$, and $U \in \mathcal{B}$, map

$$\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V) \quad \rho_U^U = id_{\mathcal{F}(U)}$$

And for $U \subseteq V \subseteq W$, we have

$$\rho_W^U = \rho_V^U \circ \rho_W^V$$

- A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves of sets on $\mathcal{B}$ is a rule which assigns to each element $U \in \mathcal{B}$ a map of sets

$$\varphi : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

compatible with restriction maps.

With the definition above, for the sheaf on $\mathcal{B}$ we have

$$\mathcal{F}_x = colim_{U \in \mathcal{B}, x \in U} \mathcal{F}(U)$$

The limit is directed. Because of the neighborhoods around x is a directed system.

Sheaves

Definition 1.4.2.2 Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X :

- A **sheaf of sets** on $\mathcal{B}$ is a presheaf of sets on $\mathcal{B}$ which satisfies the following additional property: Given any $U \in \mathcal{B}$, and any covering $U = \cup_i U_i$, with $U_i \in \mathcal{B}$, and any coverings $U_i \cap U_j = \cup_{k \in I_{ij}} U_{ijk}$, and the sheaf condition:

For any collection of sections $s_i \in \mathcal{F}(U_i), i \in I$ such that $\forall i, j \in I, \forall k \in I_{ij}$,

$$s_i|_{U_{ijk}} = s_j|_{U_{ijk}}$$

Then, there exists a unique section $s \in \mathcal{F}(U)$ such that

$$s_i = s|_{U_i}, i \in I$$

- A morphism of sheaves of sets on $\mathcal{B}$ is simply a morphism of presheaves of sets.

Remark

Extension of Sheaf

- ◇ **Lemma 1.4.2.3** Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X .
Let $\mathcal{F}$ be a sheaf of sets on $\mathcal{B}$. There exists a unique sheaf of sets $\mathcal{F}^{ext}$ on X such that

$$\mathcal{F}^{ext}(U) = \mathcal{F}(U); U \in \mathcal{B}$$

and compactibly with the restriction mappings.

◦ **Proof ...**

□

★ **Corollary 1.4.2.4** Let X be a topological space, $\mathcal{B}$ a basis for the topology on $\mathcal{B}$. Denote $Sh(\mathcal{B})$ for the category of sheaves on $\mathcal{B}$, there is an equivalence of categories:

$$Sh(X) \rightarrow Sh(\mathcal{B})$$

which assigns to a sheaf on X its restriction to the members of $\mathcal{B}$.

◦ **Proof Omitted.**

□

Algebraic Structure

Definition 1.4.2.5 Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X . Let (C, F) be a type of algebraic structure:

- A **presheaf** $\mathcal{F}$ with values in C on $\mathcal{B}$ is a rule which assign to each $U \in \mathcal{B}$ an object $\mathcal{F}(U)$ of C and to each inclusion $V \subseteq U$ of elements of $\mathcal{B}$, a morphism

$$\rho_V^U : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$$

in C with $\rho_U^U := id_{\mathcal{F}(U)}$ for all $U \in \mathcal{B}$.

Whenever $W \subseteq V \subseteq U$ in $\mathcal{B}$ we have $\rho_W^U = \rho_W^V \circ \rho_V^U$.

- A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves with values in C on $\mathcal{B}$ is a rule which assigns to each element $U \in \mathcal{B}$ a morphism of algebraic structures

...

- Given a presheaf $\mathcal{F}$ with values in C on $\mathcal{B}$, we say that

$$U \mapsto \mathcal{F}(U)$$

is the underlying presheaf of sets;

- A **sheaf** $\mathcal{F}$ with values in C on $\mathcal{B}$ is a presheaf with values in C on $\mathcal{B}$ whose underlying presheaf of sets is a sheaf.

◇ **Lemma 1.4.2.6** Let X be a topological space, (C, F) be a type of algebraic structure.

Let $\mathcal{B}$ be a basis for the topology on X , let $\mathcal{F}$ be a sheaf with values in C on $\mathcal{B}$. There exists a unique sheaf

$$\mathcal{F}^{ext}$$

with values in C on X such that

$$\mathcal{F}^{ext}(U) = \mathcal{F}(U); U \in \mathcal{B}$$

compatibly with the restriction mappings.

◦ **Proof**

$$\mathcal{F}^{ext}(U) := \text{colim}_{\mathcal{U}}$$

□

◇ **Lemma 1.4.2.7** Let X be a topological space, $\mathcal{B}$ be a basis for the topology on X .

Let (C, F) be a type of algebraic structure, denote $Sh(\mathcal{B}, C)$ the category of sheaves with values in C on $\mathcal{B}$, there is an equivalence of categories:

...

which assigns to a sheaf on X its restriction to the member of $\mathcal{B}$.

◦ **Proof ...**

□

Presheaf of Modules on Bases

Definition 1.4.2.8 Let X be a topological space, $\mathcal{B}$ a basi for the topology on X , $\mathcal{O}$ a presheaf of rings on $\mathcal{B}$:

- A **presheaf** of $\mathcal{O}$ -modules $\mathcal{F}$ on $\mathcal{B}$ is a presheaf of abelian groups on $\mathcal{B}$, together with a morphism of presheaves of sets

$$\mathcal{O} \times \mathcal{F} \rightarrow \mathcal{F}$$

Such that: $\mathcal{O}(U) \times \mathcal{F}(U) \rightarrow \mathcal{F}(U)$ turns the group into $\mathcal{O}(U)$ -module;

- A morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ of presheaves of $\mathcal{O}$ -modules on $\mathcal{B}$ is a morphism of abelian presheaves on $\mathcal{B}$ which induces an $\mathcal{O}(U)$ -module homomorphism

$$\mathcal{F}(U) \rightarrow \mathcal{G}(U)$$

- Suppose thaat $\mathcal{O}$ is a sheaf of rings on $\mathcal{B}$. A **sheaf** $\mathcal{F}$ of $\mathcal{O}$ -modules on $\mathcal{B}$ is a presheaf of $\mathcal{O}$ -modules on $\mathcal{B}$, whose underlying presheaf of abelian groups is a sheaf.

Then, define the stalk:

-

$$\mathcal{F}_x := \text{colim}_{x \in U \in \mathcal{B}} \mathcal{F}(U)$$

It is a $\mathcal{O}_x$ -module.

◇ **Lemma 1.4.2.9** (Extension of Sheaf on Bases) Let X be a topological space, with basis $\mathcal{B}$. Let $\mathcal{O}$ a sheaf of rings on $\mathcal{B}$.

Let $\mathcal{F}$ be a sheaf of $\mathcal{O}$ -modules on $\mathcal{B}$, $\mathcal{O}^{ext}$ be the sheaf of rings on X extending $\mathcal{O}$ and let $\mathcal{F}^{ext}$ be the abelian sheaf on X extending $\mathcal{F}$. Therer exists a canonical map

$$\mathcal{O}^{ext} \times \mathcal{F}^{ext} \rightarrow \mathcal{F}^{ext}$$

which agrees with the given map over elements of $\mathcal{B}$, which endows $\mathcal{F}^{ext}$ with the structure of an $\mathcal{O}^{ext}$ -module.

◦ **Proof ...**

□

★ **Corollary 1.4.2.10** Let

◦ **Proof ...**

□

Sheaf under Continuous Map

1.4.3 Gluing Sheaves

◇ **Lemma 1.4.3.1** (Pasting of Morphisms) Let X be a topological space and $X = \cup U_i$ be an open covering, and $\mathcal{F}, \mathcal{G}$ are sheaves of sets on X . Given a collection of morphisms

$$\varphi_i : \mathcal{F}|_{U_i} \rightarrow \mathcal{G}|_{U_i}$$

with $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, then there exists a unique map of sheaves

$$\varphi : \mathcal{F} \rightarrow \mathcal{G} \quad \varphi|_{U_i} = \varphi_i$$

◦ **Proof** Define:

$$\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U) \quad s \mapsto \varphi(U)(s)$$

where $\varphi(U)(s)$ is determined by $(\varphi(U \cap U_i)(s|_{U \cap U_i}))$.

$$\varphi(U)(s)|_{U \cap U_i} = \varphi_i(U \cap U_i)(s|_{U \cap U_i})$$

We should verify the condition for pasting axiom.

$$\begin{aligned} \varphi_i(U \cap U_i)(s|_{U \cap U_i})|_{U_i \cap U_j} &= \varphi_i(U \cap U_i \cap U_j)(s|_{U_i \cap U_j}) \\ &= \varphi_j(U \cap U_i \cap U_j)(s|_{U_i \cap U_j}) \\ &= \varphi_j(U \cap U_j)(s|_{U \cap U_j})|_{U_i \cap U_j} \end{aligned}$$

And φ is a sheaf map: Let $U \subseteq V \subseteq X$,

$$(\varphi(U)(s))|_V = \varphi_V(s|_V)$$

Because of :

...

□

◇ **Lemma 1.4.3.2** (Pasting of Sheaf) Let X be a topological space.

◦ **Proof** ...

□

◇ **Lemma 1.4.3.3** (Pasting of Sheaf of Abelian Groups)

◦ **Proof** ...

□

◇ **Lemma 1.4.3.4** (Pasting of Sheaf of Algebraic Structures)

◦ **Proof** ...

□

◇ **Lemma 1.4.3.5** (Pasting of Sheaf of Modules)

◦ **Proof** ...

□

Chapter 2 Sheaves of Modules

2.1 Fundamentals

2.1.1 Sheaves of Modules

Sheaves of Modules is an Abelian Category

Definition 2.1.1.1 Let $(X, \mathcal{O}_X)$ be a ringed space (1.3.0.1); Let $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$ -modules (1.1.2.10).

- Let $\varphi, \psi : \mathcal{F} \rightarrow \mathcal{G}$ be $Mod(\mathcal{O}_X)$ -morphisms. Then, define:

$$\varphi + \psi : \mathcal{F} \rightarrow \mathcal{G} \quad (\varphi + \psi)(U) = \varphi(U) + \psi(U)$$

It is clearly, that $(\varphi + \psi)$ is a morphism of sheaves of $\mathcal{O}_X$ -modules.

- Let $f : \mathcal{F} \rightarrow \mathcal{G}, g : \mathcal{G} \rightarrow \mathcal{H}$ are $\mathcal{O}_X$ -module homomorphisms, then their composition

$$g \circ f$$

is $\mathcal{O}_X$ -module.

- $Mod(\mathcal{O}_X)$ has a bilinear form

$$Mor(\mathcal{G}, \mathcal{H}) \times Mor(\mathcal{F}, \mathcal{G}) \rightarrow Mor(\mathcal{F}, \mathcal{H})$$

$$(g, f) \mapsto g \circ f$$

- Final and Initial Object(Zero Object) of $Mod(\mathcal{O}_X)$ is $0 : U \mapsto 0$.
- $\diamond$ **Lemma 2.1.1.2** Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be the morphism of $\mathcal{O}_X$ -modules. Then, the followings are equivalent:
 - φ is zero;
 - φ factors through 0;
 - φ is zero on sections over each open U ;
 - $\varphi_x = 0, x \in X$.

◦ **Proof** Here: 1.1.4.2

□

-
- Finite Direct Sum, Finite Product

$$\mathcal{F} \oplus \mathcal{G} = \mathcal{F} \times \mathcal{G}$$

Thus,

♠ **Proposition 2.1.1.3** $Mod(\mathcal{O}_X)$ is an additive category

◦ **Proof** Omitted, with maps (i, j, p, q) .

□

Kernel, Cokernel
Definition 2.1.1.4

- Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of $\mathcal{O}_X$ -modules. Then,

$$\text{Ker}(\varphi)(U) = \{s \in \mathcal{F}(U) \mid \varphi(s) = 0 \in \mathcal{G}(U)\}$$

for all open $U \subseteq X$.

- Define the cokernel: $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be $\mathcal{O}_X$ -modules morphism.

$$\text{Coker}(\varphi) := (U \mapsto \mathcal{G}(U)/\varphi(\mathcal{F}(U)))^\#$$

They are categoricial kernel, cokernel, with decomposition

$$\begin{array}{c} \bullet \varphi \circ \alpha = 0 \Leftrightarrow \begin{array}{ccccc} & \text{Ker}(\varphi) & \xrightarrow{\quad} & \mathcal{F} & \xrightarrow{\varphi} & \mathcal{G} \\ & \uparrow \alpha & \nearrow & & & \\ & \mathcal{H} & & & & \end{array} \\ \\ \bullet \beta \circ \varphi = 0 \Leftrightarrow \begin{array}{ccccc} \mathcal{F} & \xrightarrow{\varphi} & \mathcal{G} & \longrightarrow & \text{Coker}(\varphi) \\ & & \searrow \beta & \downarrow & \\ & & & \mathcal{H} & \end{array} \end{array}$$

The cokernel is defined as $\text{Coim}(\varphi) = \text{Coker}(\text{Ker} \varphi \rightarrow \mathcal{F})$.

Exact Sequence

- ◇ **Lemma 2.1.1.5** Let $(X, \mathcal{O}_X)$ be a ringed space. The category $\text{Mod}(\mathcal{O}_X)$ is an abelian category. Moreover, a complex

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

is exact at $\mathcal{G}$ iff

$$\mathcal{F}_x \longrightarrow \mathcal{G}_x \longrightarrow \mathcal{H}_x$$

is exact at $\mathcal{G}_x$.

 ◦ **Proof**

- Firstly, $\text{Mod}(\mathcal{O}_X)$ is additive. Thus there exists a morphism

$$\alpha : \text{Coim}(f) \rightarrow \text{Im}(f)$$

With universal property

$$\mathcal{F} \longrightarrow \text{Coim}(f) \xrightarrow{\alpha} \text{Im}(f) \longrightarrow \mathcal{G}$$

With lemma:1.1.4.2, the functor $(-)_x : \text{Mod}(\mathcal{O}_X) \rightarrow \mathcal{O}_{X,x} - \text{Mod}$ is exact. Thus

$$(\text{coim}(f))_x = \text{coim}(f_x)$$

$$(\text{Im}(f))_x = \text{im}(f_x)$$

α_x is an isomorphism, α is an isomorphism, thus is an abelian category.

- Omitted.

□

Product, Sums, Limits, Colimits**Definition 2.1.1.6**

- Given any set I and for each $i \in I$ a $\mathcal{O}_X$ -module we can form the product

$$\prod_{i \in I} F_i$$

which is the sheaf that associates to each open U the product of the modules $\mathcal{F}_i(U)$;

- Given any set I and for each $i \in I$ a $\mathcal{O}_X$ -module we can form the direct sum

$$\bigoplus_{i \in I} F_i = (U \mapsto \bigoplus_{i \in I} \mathcal{F}_i(U))^+$$

which is the categorical coproduct.

Using these we conclude that all limits and colimits exist in $Mod(\mathcal{O}_X)$.

◇ **Lemma 2.1.1.7** Let $(X, \mathcal{O}_X)$ be a ringed space:

- All limits exist in $Mod(\mathcal{O}_X)$, same as corresponding limits of presheaves;
- All colimits exist in $Mod(\mathcal{O}_X)$, which are sheafification of colimits of presheaves;
- Filtered Limits are exact;
- Finite direct sums are the same as corresponding finite direct sums of presheaves of $\mathcal{O}_X$ -modules.

◦ **Proof**

-
-
-
-

□

◇ **Lemma 2.1.1.8** Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces:

- $f_* : Mod(\mathcal{O}_X) \rightarrow Mod(\mathcal{O}_Y)$ is left exact, commutes with all limits;
- $f^* : Mod(\mathcal{O}_Y) \rightarrow Mod(\mathcal{O}_X)$ is right exact, commutes with all colimits;
- Pullback $f^{-1} : Ab(Y) \rightarrow Ab(X)$ on abelian sheaves is exact.

◦ **Proof**

- Notice that: f_* and f^* are adjoint pairs;
- We could check this stalkwise.

□

◇ **Lemma 2.1.1.9** Let $j : U \rightarrow X$ be an open immersion of topological spaces. The functor $j_! : AB(U) \rightarrow Ab(X)$ is exact.

◦ **Proof ...**

□

◇ **Lemma 2.1.1.10** Let $(X, \mathcal{O}_X)$ be a ringed space, I be a set. For $i \in I$, let $\mathcal{F}_i$ be a sheaf of $\mathcal{O}_X$ -modules. For quasi-compact open $U \subseteq X$, the map

$$\bigoplus_{i \in I} \mathcal{F}_i(U) \rightarrow (\bigoplus_{i \in I} \mathcal{F}_i)(U)$$

is bijective.

◦ **Proof ...**

□

Sections

Definition 2.1.1.11

Let $(X, \mathcal{O}_X)$ be a topological space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. There is a unique map of $\mathcal{O}_X$ -modules

$$\mathcal{O}_X \rightarrow \mathcal{F} \quad f \mapsto fs$$

associated to s .

The notation above signified that a local section f of $\mathcal{O}_X$. i.e. $f \in \Gamma(U, \mathcal{O}_X)$, is mapped to the multiplication of f with restriction of s to U .

$$f \mapsto f \cdot s|_U,$$

Conversely, any $(\mathcal{O}_X$ -module) map $\varphi : \mathcal{O}_X \rightarrow \mathcal{F}$ gives rise to a section $s = \varphi(1)$ such that φ is the morphism associated to s .

$$\varphi : \mathcal{O}_X \rightarrow \mathcal{F} \quad f \mapsto f \cdot s$$

$$\text{Hom}_{\mathcal{O}_X}(\mathcal{O}_X, \mathcal{F}) \cong \Gamma(X, \mathcal{F})$$

We could use a family of maps $\mathcal{O}_X \rightarrow \mathcal{F}$ to construct some of sheaf (sheaf generated by global sections)

Supports of Modules and Sections

Definition 2.1.1.12 Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules:

- The **support** of $\mathcal{F}$ is the set of points $x \in X$ such that $\mathcal{F}_x \neq 0$

$$\text{supp}(\mathcal{F}) := \{x \in X | \mathcal{F}_x \neq 0\}$$

- Let $s \in \Gamma(X, \mathcal{F})$ be a global section. The **support** of s is the set of points $x \in X$ such that $s_x \in \mathcal{F}_x$ of s is not zero

$$\text{supp}(s) := \{x \in X | s_x \neq 0\}$$

Let $s \in \Gamma(U, F)$ be a local section. The **support** of s is the set of points $x \in X$ such that $s_x \in \mathcal{F}_x$ of s is not zero

$$\text{supp}(s) = \{x \in U \mid s_x \neq 0\}$$

◇ **Lemma 2.1.1.13** Let $(X, \mathcal{O}_X)$ be a ringed space, and $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules, $U \subseteq X$ open

- $\text{supp}(s)$ is closed in U , $s \in \Gamma(U, F)$;
- $\text{supp}(fs) \subseteq \text{supp}(f) \cap \text{supp}(s)$;
- $\text{supp}(s + s') \subseteq \text{supp}(s) \cup \text{supp}(s')$;
- $\text{supp}(\mathcal{F}) = \bigcup_{s \in \Gamma(U, \mathcal{F}), U \in \text{Open}(X)} \text{supp}(s)$;
- If $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of $\mathcal{O}_X$ -modules, then

$$\text{supp}(\varphi(s)) \subseteq \text{supp}(s); s \in \mathcal{F}(U)$$

◦ **Proof**

- Conversely, $X - \text{supp}(s)$ is open;
- $X - \text{supp}(fs) \supseteq (X - \text{supp}(f)) \cup (X - \text{supp}(s))$;
- $X - \text{supp}(s + s') \supseteq (X - \text{supp}(s)) \cap (X - \text{supp}(s'))$;
- Omitted;
- Omitted;

□

◇ **Lemma 2.1.1.14** Let X be a topological space. The support of a sheaf of rings is closed.

◦ **Proof ...**

□

A Little Discussion: Quotient

Definition 2.1.1.15 The **quotient** is defined as

$$(U \mapsto \mathcal{F}(U)/\mathcal{G}(U))^+$$

Remark The quotient of presheaf $(U \mapsto \mathcal{F}(U)/\mathcal{G}(U))$ is not a sheaf. But it is **separated**: For $U = \bigcup U_i$

$$0 = \bar{s} \mid U_i \in \mathcal{F}(U_i)/\mathcal{G}(U_i) \Leftrightarrow s \mid U_i \in \mathcal{G}(U_i) \Leftrightarrow s \in \mathcal{G}(U) \Leftrightarrow 0 = \bar{s} \in \mathcal{F}(U)/\mathcal{G}(U)$$

2.1.2 Morphisms

Closed Immersions and Abelian Sheaves

◇ **Lemma 2.1.2.1** Let X be a topological space. $Z \subseteq X$ be a closed subset. Denote

$$i : Z \rightarrow X$$

the inclusion map. The functor $i_* : \text{Ab}(Z) \rightarrow \text{Ab}(X)$ is

- Exact;
- Fully faithful;
- With essential image exactly those abelian sheaves whose support is contained in Z .

And: i^{-1} is left inverse to i_* .

◦ **Proof ...**

□

Definition 2.1.2.2 Let X be a topological space. Let $Z \subseteq X$ be a closed subset. Let $\mathcal{F}$ be an abelian sheaf on X , $U \subseteq X$ open, define the sheaf:

$$\mathcal{H}_Z(U) := \{s \in \mathcal{F}(U) \mid \text{supp}(s) \subseteq Z \cap U\}$$

Then, $\mathcal{H}_Z$ is an abelian subsheaf of $\mathcal{F}$, which is the largest subsheaf such that

$$\text{supp}(\mathcal{H}_Z) \subseteq Z$$

We may view $\mathcal{H}_Z(\mathcal{F})$ as an abelian sheaf on Z . With left exact functor:

$$\text{Ab}(X) \rightarrow \text{Ab}(Z) \quad \mathcal{F} \mapsto \mathcal{H}_Z(\mathcal{F})$$

◇ **Lemma 2.1.2.3** ...

◦ **Proof ...**

□

A Canonical Exact Sequence

Bilinear Maps

Tensor Product

Duals

Symmetric and Exterior Power

Internal Hom

The Annihilator of Sheaf of Modules

Definition 2.1.2.4 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. There is a canonical map of sheaves of $\mathcal{O}_X$ -modules

$$\mathcal{O}_X \rightarrow \text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F}) \quad \mathcal{O}_X(U) \ni f \mapsto \varphi_f : s \mapsto s \cdot f$$

Then, define the **annihilator**

$$\text{Ann}_{\mathcal{O}_X} := \text{Ker}(\mathcal{O}_X \rightarrow \text{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{F}))$$

◇ **Lemma 2.1.2.5** We have:

$$(\text{Ann}_{\mathcal{O}_X}(\mathcal{F}))_x \subseteq \text{Ann}_{\mathcal{O}_{X,x}}(\mathcal{F}_x)$$

◦ **Proof ...**

□

Furthermore, we have

♠ **Proposition 2.1.2.6**

◦ **Proof ...**

□

2.2 Classification of Modules

2.2.1 Sheaf Generated by Global Sections

Definition 2.2.1.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ generated by global sections if there exists a set I , and global sections $s_i \in \Gamma(X, \mathcal{F}), i \in I$ such that the map

$$\bigoplus_{i \in I} \mathcal{O}_X \rightarrow \mathcal{F}$$

is surjective. In this case we say the sections s_i generate $\mathcal{F}$.

◇ **Lemma 2.2.1.2** Let $(X, \mathcal{O}_X)$ be a ringed space, and $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. Let I be a set. $s_i \in \Gamma(X, \mathcal{F}), i \in I$ be global sections.

The section s_i generate $\mathcal{F}$ iff for all $x \in X$, the elements $s_{i,x} \in \mathcal{F}_x$ generate the $\mathcal{O}_{X,x}$ -module $\mathcal{F}_x$.

◦ **Proof**

- $\Rightarrow$
- $\Leftarrow$

□

◇ **Lemma 2.2.1.3** Let $(X, \mathcal{O}_X)$ be a ringed space, let $\mathcal{F}, \mathcal{G}$ be sheaves of $\mathcal{O}_X$ -modules, then : If $\mathcal{F}, \mathcal{G}$ are generated by global sections, so is $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$.

◦ **Proof ...**

□

Smallest Subsheaf

◇ **Lemma 2.2.1.4** Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. I be a set.

Let $s_i, i \in I$ be a family of local sections $s_i \in \mathcal{F}(U_i)$.

Then, there exists a unique smallest subsheaf of $\mathcal{O}_X$ -modules $\mathcal{G}$ such that s_i corresponds to a local section of $\mathcal{G}$.

◦ **Proof ...**

□

Definition 2.2.1.5 ...

◇ **Lemma 2.2.1.6** ...

◦ **Proof ...**

□

2.2.2 Modules locally Generated by Sections

Definition 2.2.2.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ modules.

We say $\mathcal{F}$ is locally generated by sections if for every $x \in X$, there exists an open neighborhood U of x such that $\mathcal{F}|_U$ is globally generated as a sheaf of $\mathcal{O}_U$ -modules.

$$\bigoplus_{i \in I} \mathcal{O}_i \longrightarrow \mathcal{F}|_U \longrightarrow 0$$

◇ **Lemma 2.2.2.2** Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{G}$ is generated by sections if $\mathcal{G}$ is locally generated by sections.

◦ **Proof ...**

□

2.2.3 Modules of Finite Type

Definition 2.2.3.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules.

We say that $\mathcal{F}$ is of **finite type** if for every $x \in X$, there exists an open neighborhood U such that $\mathcal{F}|_U$ is generated by finitely many sections.

Equivalently, the following sequence is exact

$$\bigoplus_{i=1}^n \mathcal{O}_U \rightarrow \mathcal{F}|_U \rightarrow 0$$

◇ **Lemma 2.2.3.2** Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. The pullback $f^*\mathcal{G}$ of a finite type $\mathcal{O}_Y$ -module is a finite type $\mathcal{O}_X$ -module.

◦ **Proof ...**

□

◇ **Lemma 2.2.3.3** Let X be a ringed space. Then,

- The image of a morphisms of $\mathcal{O}_X$ -modules of finite type is of finite type;
- Let $0 \rightarrow \mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \mathcal{F}_3 \rightarrow 0$ be a short exact sequence of $\mathcal{O}_X$ -modules. If $\mathcal{F}_1, \mathcal{F}_3$ are of finite type, so is $\mathcal{F}_2$.

◦ **Proof ...**

□

◇ **Lemma 2.2.3.4** ...

◦ **Proof ...**

□

◇ **Lemma 2.2.3.5** ...

◦ **Proof ...**

□

2.2.4 Quasi-Coherent Modules

Definition 2.2.4.1 Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules.

We say that $\mathcal{F}$ is **quasi-coherent sheaf** of $\mathcal{O}_X$ -module if for every point $x \in X$, there exists an open neighborhood $U \subseteq X$ such that

$$\mathcal{F}|_U = \text{Coker}(\oplus_{j \in J} \mathcal{O}_U \rightarrow \oplus_{i \in I} \mathcal{O}_U)$$

Take the notation of quasi-coherent $\mathcal{O}_X$ -modules by $QCoh(\mathcal{O}_X)$.

Equivalently, $\mathcal{F}|_U$ has a presentation of the form

$$\oplus_{j \in J} \mathcal{O}_U \rightarrow \oplus_{i \in I} \mathcal{O}_U \rightarrow \mathcal{F}|_U \rightarrow 0$$

In other word:

- For every point $x \in X$, there exists an open neighborhood $U \subseteq X$, and $\mathcal{F}|_U$ is generated by local sections;
- For a suitable choice of these sections, the kernel of associated surjection is also generated by local sections.

Sum

◇ **Lemma 2.2.4.2** Let $(X, \mathcal{O}_X)$ be a ringed space. The direct sum of two two quasi-coherent $\mathcal{O}_X$ -modules is a quasi-coherent $\mathcal{O}_X$ -modules.

- **Proof** Take the disjoint union of indexes $I \sqcup I', J \sqcup J'$. □

Pullback

◇ **Lemma 2.2.4.3** Let $f : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ be a morphism. The **pullback** $f^*\mathcal{G}$ of quasi-coherent $\mathcal{O}_Y$ -module is quasi-coherent.

- **Proof** Firstly, consider the following diagram:

$$\oplus_{i \in I} \mathcal{O}_U \longrightarrow \oplus_{j \in J} \mathcal{O}_U \longrightarrow \mathcal{G}|_U \longrightarrow 0$$

Apply the operation $f^* = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1} - \Rightarrow f^*|_U = \mathcal{O}_X|f^{-1}(U) \otimes_{f^{-1}\mathcal{O}_Y|U} f^{-1}\mathcal{O}_U$, exactness

$$\oplus_{i \in I} \mathcal{O}_X|f^{-1}(U) \longrightarrow \oplus_{j \in J} \mathcal{O}_X(U)|f^{-1}(U) \longrightarrow f^*\mathcal{G}|f^{-1}(U) \longrightarrow 0$$

□

Sheaf Associated to the Module M and Ring Map α

◇ **Lemma 2.2.4.4** Let $(X, \mathcal{O}_X)$ be ringed space, $\alpha : R \rightarrow \Gamma(X, \mathcal{O}_X)$ be a ring homomorphism from a ring R into the ring of global sections on X .

Let M be an R -module. The following three constructions give canonically isomorphic sheaves of $\mathcal{O}_X$ -modules:

- Let $\pi : (X, \mathcal{O}_X) \rightarrow (\{*\}, R)$ be the morphism of ringed spaces with:
 - $\pi : X \rightarrow \{*\}$ unique map;
 - π map $\pi^\#(\{x\}) = \alpha : R \rightarrow \Gamma(X, \mathcal{O}_X)$. Set

$$\mathcal{F}_1 := \pi^*M$$

- Choose a presentation $\bigoplus_{j \in J} R \rightarrow \bigoplus_{i \in I} R \rightarrow M \rightarrow 0$, set

$$\mathcal{F}_2 = \text{Coker}(\bigoplus_{j \in J} \mathcal{O}_X \rightarrow \bigoplus_{i \in I} \mathcal{O}_X)$$

- Set $\mathcal{F}_3$ equal to the sheaf associated to the prehsheaf

$$U \mapsto \mathcal{O}_X(U) \otimes_R M$$

where: $R \rightarrow M$ by module structure, $R \rightarrow \mathcal{O}_X(U)$ the composition of α and the composition with restriction map $\Gamma(X, \mathcal{O}_X) \rightarrow \Gamma(U, \mathcal{O}_X)$.

The construction has the following properties:

- The **resulting sheaf** of $\mathcal{O}_X$ -modules

$$\mathcal{F}_M = \mathcal{F}_1 = \mathcal{F}_2 = \mathcal{F}_3$$

is quasi-coherent.

- The construction gives a functor from the category of R -modules to the category of quasi-coherent sheaves on X , which commutes with arbitrary colimits;
- For any $x \in X$, we have

$$\mathcal{F}_{M,x} = \mathcal{O}_{X,x} \otimes_R M$$

- Given any $\mathcal{O}_X$ -module $\mathcal{G}$ we have

$$\text{Mor}_{\mathcal{O}_X}(\mathcal{F}_M, \mathcal{G}) = \text{Hom}_R(M, \Gamma(X, \mathcal{G}))$$

◦ Proof

- –
-
-
- –
-
-
-

□

Definition 2.2.4.5 Such sheaf $\mathcal{F}_M$ is the sheaf associated to the module M and the ring map $\alpha : R \rightarrow \Gamma(X, \mathcal{O}_X)$.

◇ **Lemma 2.2.4.6** Let $(X, \mathcal{O}_X)$ be a ringed space, $R = \Gamma(X, \mathcal{O}_X)$.

Let M be an R -module, $\mathcal{F}_M$ be the quasi-coherent sheaf of $\mathcal{O}_X$ -modules associated to M .

If $g : (Y, \mathcal{O}_Y) \rightarrow (X, \mathcal{O}_X)$ is a morphism of ringed spaces, then $g^* \mathcal{F}_M$ is the sheaf associated to the $\Gamma(Y, \mathcal{O}_Y)$ -module $\Gamma(Y, \mathcal{O}_Y) \otimes_R M$.

◦ **Proof ...**

□

◇ **Lemma 2.2.4.7** ...

◦ **Proof ...**

□

2.2.5 Modules of Finite Representation

Definition 2.2.5.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is of **finite presentation** if for every point $x \in X$, there exists an open subset $U \subseteq X$ and $n, m \in \mathbb{N}$ such that

$$\mathcal{F}|_U = \text{Coker}(\oplus_{j=1}^m \mathcal{O}_U \rightarrow \oplus_{i=1}^n \mathcal{O}_U)$$

Equivalently, the following diagram

$$\oplus_{j=1}^m \mathcal{O}_U \rightarrow \oplus_{i=1}^n \mathcal{O}_U \rightarrow \mathcal{F}|_U \rightarrow 0$$

is exact.

◇ **Lemma 2.2.5.2** Let $(X, \mathcal{O}_X)$ be a ringed space Then: Any $\mathcal{O}_X$ -module of finite presentation is quasi-coherent.

◦ **Proof ...** □

2.2.6 Coherent Modules

Definition 2.2.6.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. We say that $\mathcal{F}$ is a coherent $\mathcal{O}_X$ -module if the following condition holds:

-
- For every open $U \subseteq X$, and every finite collection $s_i \in \mathcal{F}(U), i = 1, \dots, n$,

$$\text{Ker}(\oplus \mathcal{O}_U \rightarrow \mathcal{F}|_U)$$

is of finite type.

The category of coherent $\mathcal{O}_X$ -modules is $\text{Coh}(\mathcal{O}_X)$.

2.2.7 Locally free Sheaves

Definition 2.2.7.1 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules:

- We say $\mathcal{F}$ is **locally free** if for every point $x \in X$, there exists a set I and an open neighborhood $U \subseteq X$ such that

$$\mathcal{F}|_U = \oplus_{i \in I} \mathcal{O}_X|_U$$

as an $\mathcal{O}_X|_U$ -module;

- We say $\mathcal{F}$ is **finite locally free** if we may choose the index sets I to be finite;
- We say $\mathcal{F}$ is locally free of **rank** r if we may choose the index sets of cardinality r ;

◇ **Lemma 2.2.7.2** Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{F}$ be a sheaf of $\mathcal{O}_X$ -modules. If $\mathcal{F}$ is locally free, then it is quasi-coherent.

◦ **Proof ...** □

Pullback

◇ Lemma 2.2.7.3 ...

◦ Proof ...

□

Rank

◇ Lemma 2.2.7.4 ...

◦ Proof ...

□

◇ Lemma 2.2.7.5 Let $(X, \mathcal{O}_X)$ be a ringed space, $r \geq 0$. Let $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ be a finite locally free $\mathcal{O}_X$ -modules of rank r . Then φ is an isomorphism iff φ is surjective

◦ Proof ...

□

Locally Ringed Space

◇ Lemma 2.2.7.6 Let $(X, \mathcal{O}_X)$ be a locally ringed space, then any direct summand of a finite locally free $\mathcal{O}_X$ -module is finite locally free.

◦ Proof ...

□

◇ Lemma 2.2.7.7 Let $(X, \mathcal{O}_X)$ be locally ringed space, then $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism iff φ is injective.

◦ Proof ...

□

Remark Actually, locally free sheaves is correspondent to the vector bundle. Via Serre-Swan theorem, we have a equivalence of Categories

$$Vect(X) \cong P(C^0(X))$$

$C^0(X)$ is a local ring.

2.2.8 Flat Modules

Definition 2.2.8.1 Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is **flat** if the functor

$$- \otimes_{\mathcal{O}_X} \mathcal{F} : Mod(\mathcal{O}_X) \rightarrow Mod(\mathcal{O}_X) \quad \mathcal{G} \mapsto \mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{F}$$

is exact.

◇ Lemma 2.2.8.2 Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$ -module $\mathcal{F}$ is flat iff the stalk $\mathcal{F}_x$ is a flat $\mathcal{O}_{X,x}$ -module for all $x \in X$.

◦ Proof ...

□

Definition 2.2.8.3 Let $(X, \mathcal{O}_X)$ be a ringed space. Let $x \in X$. An $\mathcal{O}_X$ -module $\mathcal{F}$ is **flat at x** if $\mathcal{F}_x$ is flat $\mathcal{O}_{X,x}$ -module.

Pullback

◇ **Lemma 2.2.8.4** ...

◦ **Proof** ...

□

Filtered Colimit

2.2.9 Constructible Sheaves of Sets

2.2.10 Flat Morphisms of Ringed Spaces

2.2.11 Invertible Modules

Definition 2.2.11.1 Let $(X, \mathcal{O}_X)$ be a ringed space. An **invertible** $\mathcal{O}_X$ -module is a sheaf of $\mathcal{O}_X$ -modules $\mathcal{L}$ such that the functor

$$Mod(\mathcal{O}_X) \rightarrow Mod(\mathcal{O}_X) \quad \mathcal{F} \mapsto \mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{F}$$

is an equivalence of categories.

We say $\mathcal{L}$ is **trivial** if its isomorphic to $\mathcal{O}_X$, as an $\mathcal{O}_X$ -modules.

◇ **Lemma 2.2.11.2** Let $(X, \mathcal{O}_X)$ be a ringed space. Let $\mathcal{L}$ be an $\mathcal{O}_X$ -module. Equivalent are:

- $\mathcal{L}$ is invertible;
- There exists an $\mathcal{O}_X$ -module $\mathcal{N}$ such that $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{N} \cong \mathcal{O}_X$
where $\mathcal{L}$ is locally a direct summand of a finite free $\mathcal{O}_X$ -module and the module $\mathcal{N}$ is isomorphic to $Hom_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$.

$$\mathcal{O}_X^n|U \cong \mathcal{L}|U \oplus \mathcal{L}'|U \quad \mathcal{N} \cong Hom_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$$

◦ **Proof**

• $\Rightarrow$

• $\Leftarrow$

□

◇ **Lemma 2.2.11.3** ...

◦ **Proof** ...

□

Tensor Power

Definition 2.2.11.4 Let $(X, \mathcal{O}_X)$ be a ringed space. Given an invertible sheaf $\mathcal{L}$ on X and $n \in \mathbb{Z}$. We define the n th tensor power:

$$\mathcal{L}^{\otimes n} := \begin{cases} \mathcal{O}_X & n = 0 \\ \text{Hom}_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X) & n = -1 \\ \underbrace{\mathcal{L} \otimes_{\mathcal{O}_X} \cdots \otimes_{\mathcal{O}_X} \mathcal{L}}_{n \text{ times}} & n > 1 \\ \underbrace{\mathcal{L}^{-1} \otimes_{\mathcal{O}_X} \cdots \otimes_{\mathcal{O}_X} \mathcal{L}}_{n \text{ times}} & n < -1 \end{cases}$$

Associated Graded Ring

◇ **Lemma 2.2.11.5** ...

◦ **Proof** ...

□

Definition 2.2.11.6 Let $(X, \mathcal{O}_X)$ be a ringed space. Given an invertible sheaf $\mathcal{L}$ on X . We define the **associated graded ring** to be

$$\begin{aligned} \Gamma_*(X, \mathcal{L}) &= \bigoplus_{n \geq 0} \Gamma(X, \mathcal{L}^{\otimes n}) \\ \Gamma_*(X, \mathcal{L}, \mathcal{F}) &= \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{L}^{\otimes n}) \end{aligned}$$

Take the notation of $\Gamma_*(\mathcal{L}), \Gamma_*(\mathcal{F})$.

Equipped with the multiplication given by lemma above.

Picard Group

◇ **Lemma 2.2.11.7** Let $(X, \mathcal{O}_X)$ be a ringed sapce. There exists a st of invertible modules $\{\mathcal{L}_i\}_{i \in I}$ such that: Each invertible module on X is isomorphic to exactly one of the $\mathcal{L}_i$.

◦ **Proof** ...

□

Definition 2.2.11.8 Let $(X, \mathcal{O}_X)$ be a ringed space. The **Picard group** $\text{Pic}(X)$ is the abelian group, whose elements are isomorphism classes of invertible $\mathcal{O}_X$ -modules, with addition corresponding to tensor products.

◇ **Lemma 2.2.11.9** (Local Inversability) Let X be a ringed space. Assume that each stalk $\mathcal{O}_{X,s}$ is a local ring with m_s . Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$ -module. For any section $s \in \Gamma(X, \mathcal{L})$, the set

...

is open in X .

The map $s : \mathcal{O}_{X_s} \rightarrow \mathcal{L}|_{X_s}$ is an isomorphism, and there exists a section $s' \in \mathcal{L}^{\otimes -1}|_{X_s}$ such that

$$s'(s|_{X_s}) = 1$$

◦ **Proof ...**

□

Geometry of Invertible Module: Line Bundle

Remark The geometric correspondnece of invertible module is the line bundle: Which is a fibre bundle with fibres of lines. In K-theory:

- Addition $\Leftrightarrow$ Direct Sum;
- Multiplication $\Leftrightarrow$ Tensor Product.

2.3 Other Constructions

2.3.1 Konzul Complexes

Definition 2.3.1.1 Let X be a ringed space, and $\varphi : \epsilon \rightarrow \mathcal{O}_X$ be $\mathcal{O}_X$ -module map.

The **Konzul complex** $K_*(\varphi)$ associated to φ is the sheaf of commutative differential graded algebras defined as follows:

- The underlying graded algebra is the exterior algebra

$$K_*(\varphi) = \wedge(\epsilon)$$

- Differentials

$$d : K_*(\varphi) \rightarrow K_*(\varphi)$$

is the unique derivation such that $d(e) = \varphi(e)$ for all local section $e \in \epsilon = K_1(\varphi)$.

Precisely,

...

Definition 2.3.1.2 Let X be a ringed space, $f_1, \dots, f_n \in \Gamma(X, \mathcal{O}_X)$. Then, the **Konzul complex on** $f_1, \dots, f_n$ is the Konzul complex associated to $(f_1, \dots, f_n) : \mathcal{O}_X^{\oplus n} \rightarrow \mathcal{O}_X$.

♠ **Proposition 2.3.1.3** (Algebraic Poincare's Lemma)

$$dd\omega = 0$$

◦ **Proof ...**

□

2.3.2 Rank and Determinant

K-Groups

Now, consider the zeroth Grothendieck group

$$K_0(\text{Vect}(X))$$

where $Vect(X) = \{[\epsilon] | \epsilon \text{ is finite locally free}\}$, and K_0 is the abelian group generated by this set.

$$[\epsilon] = [\epsilon'] + [\epsilon''] \Leftrightarrow \text{Exists } 0 \rightarrow \epsilon' \rightarrow \epsilon \rightarrow \epsilon'' \rightarrow 0$$

With an exact sequence, we have local direct-sum composition via

...

Rank

Definition 2.3.2.1 Let ϵ be a finite locally free $\mathcal{O}_X$ -module ϵ . The rank is a locally constant function

$$rank_\epsilon : X \rightarrow \mathbb{Z}_{\geq 0} \quad x \mapsto rank_{\mathcal{O}_{X,x}}(\epsilon_x)$$

There exists a homomorphism

$$K_0(Vect(X)) \rightarrow Map_{cont}(X, \mathbb{Z}) \quad [\epsilon] \mapsto rank_\epsilon$$

Here, $Map_{cont}(-, -)$ for locally constant functions.

Determinant

Definition 2.3.2.2 Let $X = \coprod X_i$. With X_i open and closed.

2.3.3 Localization

2.3.4 Modules of Differentials

2.3.5 Finite Order differential Operators

2.3.6 De Rham Complex

2.3.7 The Naive Cotangent Complex

Definition 2.3.7.1 (Polynomial Algebras)

2.4 Differential Graded Sheaves

2.4.1 Sheaves of Graded Algebras

Definition 2.4.1.1 Let $(C, \mathcal{O})$ be a ringed site. A **sheaf of graded $\mathcal{O}$ -algebras** or a **sheaf of graded algebras on $(C, \mathcal{O})$** is given by

- A family $\mathcal{A}^n, n \in \mathbb{Z}$;
- $\mathcal{O}$ -bilinear maps

$$\begin{aligned} \mathcal{A}^n \times \mathcal{A}^m &\rightarrow \mathcal{A}^{n+m} \\ (a, b) &\mapsto ab \end{aligned}$$

called the **multiplication maps** with following properties:

-
-

Homomorphism

Definition 2.4.1.2 ...

2.4.2 Sheaves of Graded Modules**2.4.3 The Graded Category of Sheaves of Graded Modules****2.4.4 Tensor Product for Sheaves of Graded Modules****2.4.5 Internal Hom for Sheaves of Graded Modules****2.4.6 Localization and Sheaves of Graded Modules****2.4.7 Homotopy Category**

Definition 2.4.7.1 Let $(C, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebras on $(C, \mathcal{O})$. Let $f, g : \mathcal{M} \rightarrow \mathcal{N}$ be differential graded $\mathcal{A}$ -modules.

A **homotopy** between f and g is a graded $\mathcal{A}$ -module map $h : \mathcal{M} \rightarrow \mathcal{N}$ homogeneous of degree -1 such that

$$f - g = d_{\mathcal{N}} \circ h + h \circ d_{\mathcal{M}}$$

Cone, and Triangles

◇ **Lemma 2.4.7.2** Let $(C, \mathcal{O})$ be a ringed site. Let $\mathcal{A}$ be a sheaf of differential graded algebra

Chapter 3 Sheaves over Abelian Groups

In this chapter, we will construct the sheaves via covering space.

3.1 Definitions

Definition 3.1.0.1 A **sheaf** on X is a pair $(\mathcal{A}, \pi)$ where

- $\mathcal{A}$ is a topological space;
- $\pi : \mathcal{A} \rightarrow X$ is a local homeomorphism onto X ;
- Each fiber $\mathcal{A}_x = \pi^{-1}(x)$ is an abelian group;
- The group operations are continuous.

Definition 3.1.0.2 Take the following notation:

- Define:

$$\mathcal{A} \triangle \mathcal{A} = \{(\alpha, \beta) | \pi(\alpha) = \pi(\beta)\} \subseteq \mathcal{A} \times \mathcal{A}$$

- Continuous Operations:

$$\begin{aligned} \mathcal{A} \triangle \mathcal{A} &\rightarrow \mathcal{A} \\ (\alpha, \beta) &\mapsto \alpha - \beta \end{aligned}$$

Or equivalently, two operations:

–
–

Similarly, we have defined the

-
-

Restriction

Definition 3.1.0.3 Let $\mathcal{A}$ be a sheaf and $Y \subseteq X$. Then, the **restriction** of $\mathcal{A}$ on Y , denoted by $\mathcal{A}|Y$ is defined as:

$$\mathcal{A}(Y) = (\pi^{-1}(Y), \pi)$$

which is a sheaf on Y .

Section

Definition 3.1.0.4 Let $\mathcal{A}$ be a sheaf on X and $Y \subseteq X$ be an open subset, the **section** of $\mathcal{A}$ over Y is a map $s : Y \rightarrow \mathcal{A}$ such that

$$\pi \circ s = id_Y$$

Take the following notation $\mathcal{A}(Y) = \{s | \pi \circ s = id_Y\}$.

Some of facts:

♠ **Proposition 3.1.0.5**

- π is an open map;
- Any section of $\mathcal{A}$ over an open set is an open map;
- An element of $\mathcal{A}$ is in the range of some section over some open set;
- The set of all images of sections over open sets is a base for the topology of $\mathcal{A}$;
- For two sections $s \in \mathcal{A}(U), t \in \mathcal{A}(V)$, then

$$V = \{x \in Y | s(x) = t(x)\}$$

is open in Y .

◦ **Proof**

-
-
-

□

♠ **Proposition 3.1.0.6**

- (Pasting Lemma)
- (Identity Lemma)

◦ **Proof ...**

□

Sheafification

3.1.1 Homomorphisms Subsheaves and Quotient Sheaves

3.1.2 Cohomologies

Let $f : X \rightarrow Y$ be a given continuous map.

Definition 3.1.2.1

- Let A, B be presheaves on X, Y respectively, then an f -homomorphism $k : B \rightarrow A$ is

$$k(U) : B(U) \rightarrow A(f^{-1}(U))$$

for U open in Y and compatible with restrictions:

$$B(U) = A(f^{-1}(U))$$

$$B(V) = A(f^{-1}(V))$$

•

3.1.3 Algebraic Constructions

3.1.4 Supports

Chapter 4 Sheaf Cohomologies

4.1 Homological Properties

♣ **Example 4.1.0.1** Consider $X = \text{Spec}(F_p)$, $\mathcal{F} = \bar{F}_p$,
 $\mathcal{F}$ is inj. in $\text{Mod}(X, \mathcal{O}_X)$ but not in $\text{Sh}(X)$

title

4.1.1 Čech Cohomology

Definition 4.1.1.1 (Čech Cohomology) Take the following notion:

$$\mathcal{U} := (U_i)_{i \in I} \quad U_J := \bigcap_{j \in J} U_j, J \subseteq I$$

Consider the cochain

$$C^i := \prod_{J \subseteq I, |J|=i+1} \mathcal{F}(U_J) \quad d^j(s) := \prod_{J=\{j_0, \dots, j_{i+1}\}} \sum_{k=0}^{i+1} (-1)^k s_{j_0, \dots, \hat{j}_k, \dots, j_{i+1}} |U_J$$

Verify the cochain condition:

$$d^j \circ d^{j+1} =$$

First Cohomology

Definition 4.1.1.2 Let X be a topological space, $\mathcal{G}$ a sheaf of groups on X (**Remark:** Possibly even non-abelian)

◇ **Lemma 4.1.1.3** ...

◦ **Proof** ...

□

◇ **Lemma 4.1.1.4** Let X be a topological space, $\mathcal{H} \in \text{Ab}(X)$. Then, there exists a canonical isomorphism between set of

◦ **Proof** ...

□

Extensions

Invertible Sheaves

◇ **Lemma 4.1.1.5** (Picard Group)

◦ **Proof** ...

□

4.1.2 Locality of Cohomology**4.1.3 Mayer-Vietoris****4.1.4 Functoriality of Cohomology****4.1.5 Refinement and Čech Cohomology****4.1.6 Noetherian Space****4.2 Flasque Sheaves**

Definition 4.2.0.1 (Flasque Sheaves)

◇ **Lemma 4.2.0.2** ...

◦ **Proof** ...

□

4.3 The Leray Spectral Sequence**4.4 Base Change****4.5 Colimits****4.6 Cup Product**

Chapter 5 Sites

5.1 Definitions

5.1.1 Morphisms with Fixed Target

Definition 5.1.1.1 Let $\mathcal{C}$ be a category.

A **family of morphisms** with fixed target in $\mathcal{C}$ is given by

- an object $U \in \text{Ob}(\mathcal{C})$;
- a set I and for each $i \in I$ a morphism $U_i \rightarrow U$ of $\mathcal{C}$ with target U .

Take notation of

$$\{U_i \rightarrow U\}_{i \in I}$$

5.1.2 Sites

Definition 5.1.2.1 A **site** is given by a category $\mathcal{C}$ and a set $\text{Cov}(\mathcal{C})$ of families of morphisms with fixed target $\{U_i \rightarrow U\}_{i \in I}$ called **covering** of $\mathcal{C}$, satisfying the following axioms:

- (Isomorphism) If $V \rightarrow U$ is an isomorphism then $\{V \rightarrow U\} \in \text{Cov}(\mathcal{C})$;
- (Compositions) If $\{U_i \rightarrow U\}_{i \in I} \in \text{Cov}(\mathcal{C})$ and for each $i \in I$ we have $\{V_{ij} \rightarrow U_i\}_{j \in J_i} \in \text{Cov}(\mathcal{C})$, then

$$\{V_{ij} \rightarrow U\}_{i \in I, j \in J_i} \in \text{Cov}(\mathcal{C})$$

- (Fiber Products) If $\{U_i \rightarrow U\}_{i \in I} \in \text{Cov}(\mathcal{C})$; $(V \rightarrow U) \in \text{Mor}(\mathcal{C})$. Then,

$$\exists U_i \times_U V \in \mathcal{C}, \{U_i \times_U V \rightarrow V\}_{i \in I} \in \text{Cov}(\mathcal{C})$$

It is the generation of $\text{PSh}(\mathcal{C})$.

♣ **Example 5.1.2.2** ...